



PRECURSOR ANALYSIS REPORT: OPERATIONAL TECHNOLOGY MALWARE FRAMEWORKS BASED ON INCONTROLLER

Cybersecurity for the Operational Technology
Environment (CyOTE)

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FIGURE 3. ATTACK GRAPH [DO NOT EDIT THIS PAGE] 47

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PRECURSOR ANALYSIS REPORT: OT MALWARE FRAMEWORKS BASED ON INCONTROLLER

1. EXECUTIVE SUMMARY

The Operational Technology Malware Frameworks Based on INCONTROLLER Precursor Analysis Report leverages publicly available information about the custom Industrial Control Systems (ICS) malware framework and catalogs anomalous observables for each technique that may be employed in a deployment of the framework. This analysis is based upon the methodology of the Cybersecurity for the Operational Technology Environment (CyOTE) program.

In early April 2022, the Cybersecurity Infrastructure Security Agency (CISA), security vendors Mandiant and Dragos, as well as original equipment manufacturers (OEMs) Schneider Electric and Omron released advisories on a novel malware framework targeting assets and protocols used in critical infrastructure and operational technology (OT) environments. Named INCONTROLLER by Mandiant and PIPEDREAM by Dragos, the framework leverages several tools and targets popular programmable logic controllers (PLCs) manufactured by Omron and Schneider Electric, as well as protocols common in OT environments, including Open Platforms Communications Unified Architecture (OPC UA), Controller Development System (CODESYS), and Omron's Factory Interface Network Service (FINS). Although industry experts emphasized that the framework was not thought to have been deployed, experts also cautioned that INCONTROLLER could be scaled up to target additional OT assets not initially included in the framework.

Researchers and analysts identified **39** unique techniques associated with an adversary's capability with a total of **743** observables using MITRE ATT&CK® for Industrial Control Systems. The CyOTE program assesses observables accompanying techniques public reporting has attributed to the framework's capabilities. If observables accompanying the use of any techniques are perceived and investigated prior to the triggering event, earlier comprehension of malicious activity may take place. **Thirty-one** of the identified techniques associated with INCONTROLLER could be considered precursors before impact occurs. Analysis identified **574** observables associated with these precursor techniques, **64** of which were assessed to have an increased likelihood of being perceived prior to impact. The response and comprehension time can be reduced if the observables are identified earlier.

The information gathered in this report contributes to a library of observables tied to a repository of artifacts, data sources, and technique detection references for practitioners and developers to support the comprehension of indicators of attack. Asset owners and operators can use these products if they experience similar observables or to prepare for comparable scenarios.

2. INTRODUCTION

The [Cybersecurity for the Operational Technology Environment \(CyOTE\)](#) program developed capabilities for energy sector organizations to independently identify adversarial tactics and techniques within their operational technology (OT) environments. Led by Idaho National Laboratory (INL) under leadership of the Department of Energy (DOE) Office of Cybersecurity, Energy Security, and Emergency Response (CESER), CyOTE is a partnership with energy sector owners and operators whose goal is to tie the impacts of a cyber attack to anomalies in the OT environment to determine whether the anomalies have a malicious cyber cause.

2.1. APPLYING THE CYOTE METHODOLOGY

The CyOTE methodology, as shown in Figure 1. CyOTE Methodology, applies fundamental concepts of perception and comprehension to a universe of knowns and unknowns increasingly disaggregated into observables, anomalies, and triggering events. The program utilizes MITRE's ATT&CK® Framework for Industrial Control Systems (ICS) as a common lexicon to assess triggering events. By leveraging the CyOTE methodology with existing commercial monitoring capabilities and manual data collection, energy sector partners can understand relationships between multiple observables, which could represent a faint signal of an attack requiring investigation. CyOTE can assist organizations in prioritizing their OT environment visibility investments.

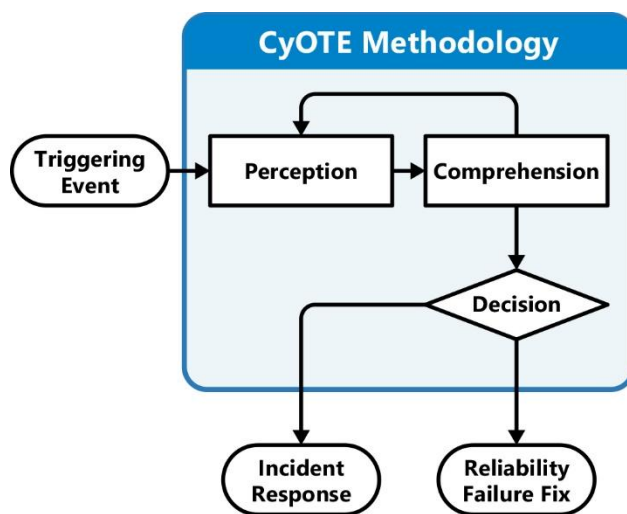


Figure 1. CyOTE Methodology

Historical case studies such as this one support continued learning through analysis of incidents that have impacted OT. This precursor analysis report is based on publicly available reports and provides examples of how key concepts in the CyOTE methodology appear in the real world, providing insights on how similar novel attacks could be detected earlier and therefore mitigated. The analysis enables OT personnel to independently identify observables associated with techniques known to be indicators of attack within OT environments. The identified observables highlight anomalous events for further investigation, which could enhance comprehension of malicious activity.

A timeline of events based on the CyOTE methodology portrays the attack-related observables associated with the precursor analysis report cyber attack. The timeline includes assessed dates, the triggering event, and comprehension of malicious activity by the organization. The point on this timeline when each technique appears is critical to the organization's ability to perceive and comprehend the associated malicious activity. Perception of techniques early in the timeline is critical, since halting those techniques will generally have greater potential to limit additional attack vectors using other techniques, defeat the cyber attack, and limit damage to operations.

Each technique has an assessed perceivability. Perceivability is a function of the number of observables and the potential for personnel to detect those observables. If a technique includes effects which personnel may detect, such as deletion or modification of system files or required user execution, then the technique would be more perceivable.

Differences in infrastructure and system configurations may present different challenges and opportunities for observable detection. For example, architecture-wide endpoint monitoring is likely to improve the perceivability of techniques which modify host files, such as the Data Destruction technique (T0809) for Inhibit Response Function and Theft of Operational Information technique (T0882) for Impact. Network monitoring and log analysis capabilities are likely to improve perceivability of techniques which create malicious network traffic, such as the Standard Application Layer Protocol technique (T0869) for Command and Control, External Remote Services technique (T0822) for Initial Access, and Connection Proxy technique (T0884) for Command and Control. Alternatively, enhancing the monitoring parameters of system files would increase the perceivability of techniques such as Data from Information Repositories technique (T0811) for Collection and the Service Stop technique (T0881) for Inhibit Response Function.

Comprehension can be further enhanced by technique artifacts created when adversaries employ certain attack techniques. The CyOTE program provides organizations with a [library of observables](#) reported in each historical case. The library can be used in conjunction with a repository of artifacts, data sources, and technique detection references for practitioners and developers to support the comprehension of indicators of attack.

2.2. BACKGROUND ON ADVERSARY CAPABILITY

On 13 April 2022, the Cybersecurity Infrastructure Security Agency (CISA), security vendors Mandiant and Dragos, as well as original equipment manufacturers (OEMs) Schneider Electric and Omron, published reports documenting a newly discovered malware framework designed to target Operational Technology (OT) protocols and specific Industrial Control Systems (ICS) components. Named INCONTROLLER¹ by Mandiant, and PIPEDREAM² by Dragos, the framework contained several tools targeting specific OT assets and protocols. All sources of the reporting indicated this framework was discovered in the wild and was not thought to have been deployed or used at its final target to cause destructive or disruptive effects.^{3,4}

Additionally, sources at Mandiant and Dragos emphasized this framework was scalable to additional OT infrastructure that use common protocols such as Modbus and Controller Development System (CODESYS), employing tools such as OMSHELL, CODECALL, TAGRUN, and ICECORE.^a OMSHELL is specifically designed to access Omron PLCs including the NX1P2, NJ501, the R88D-1SN10F servo drive, and potentially other PLCs in the NX/NJ product line.⁵ CODECALL is able to communicate with devices that use Modbus and CODESYS protocols, with a capability against specific Schneider Electric PLCs, including the M251 (M251MESE) and Modicon M258.⁶ TAGRUN targets OPC UA, a popular interoperability protocol used in OT environments that streamlines the monitoring and operation of proprietary OT devices from different manufacturers. ICECORE is malware designed to support general remote access and command and control (C2) which likely can be used to pivot from the IT environment into the OT environment.

Analysis identified **39** unique techniques associated with the tools included in the INCONTROLLER framework (**Error! Reference source not found.** 1). These attack techniques are defined according to MITRE's ATT&CK® for ICS framework.

The following analysis will detail the capabilities of INCONTROLLER in a hypothetical victim environment containing Omron and Schneider Electric controllers, and protocols OPC UA, Modbus, and CODESYS protocols, and assumes an adversary deploying INCONTROLLER will maximize the impact on the victim's environment. This analysis will primarily use Mandiant's terminology for the individual malware and associated tools, although Dragos terminology will be used to address the ASRock vulnerability (CVE-2020-15368) where necessary. Finally, already reported techniques are omitted in this report due to gaps in publicly available threat analysis.

^a This publication primarily uses Mandiant's terminology for INCONTROLLER, specifically the tools ICECORE, TAGRUN, OMSHELL, and CODECALL. Dragos refers to these tools as DUSTTUNNEL, MOUSEHOLE, BADOMEN, and EVILSCHOLAR, and the ASRock vulnerability (CVE-2020-15368) as LAZYCARGO.

Table 1. Techniques Used in the [PAR TITLE] Cyber Attack

Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Process Control	Impact
Drive-by Compromise	Change Operating Mode	Modify Program	Exploitation for Privilege Escalation	Change Operating Mode	Network Connection Enumeration	Default Credentials	Automated Collection	Commonly Used Port	Activate Firmware Update Mode	Brute Force I/O	Damage to Property
Exploit Public-Facing Application	Command-Line Interface	Module Firmware	Hooking	Exploitation for Evasion	Network Sniffing	Exploitation of Remote Services	Data from Information Repositories	Connection Proxy	Alarm Suppression	Modify Parameter	Denial of Control
Exploitation of Remote Services	Execution through API	Project File Infection		Indicator Removal on Host	Remote System Discovery	Lateral Tool Transfer	Detect Operating Mode	Standard Application Layer Protocol	Block Command Message	Module Firmware	Denial of View
External Remote Services	Graphical User Interface	System Firmware		Masquerading	Remote System Information Discovery	Program Download	I/O Image		Block Reporting Message	Spoof Reporting Message	Loss of Availability
Internet Accessible Device	Hooking	Valid Accounts		Rootkit	Wireless Sniffing	Remote Services	Man in the Middle		Block Serial COM	Unauthorized Command Message	Loss of Control
Remote Services	Modify Controller Tasking			Spoof Reporting Message		Valid Accounts	Monitor Process State		Data Destruction		Loss of Productivity and Revenue
Replication Through Removable Media	Native API						Point & Tag Identification		Denial of Service		Loss of Protection
Rogue Master	Scripting						Program Upload		Device Restart/Shutdown		Loss of Safety
Spearphishing Attachment	User Execution						Screen Capture		Manipulate I/O Image		Loss of View
Supply Chain Compromise							Wireless Sniffing		Modify Alarm Settings		Manipulation of Control
Transient Cyber Asset									Rootkit		Manipulation of View
Wireless Compromise									Service Stop		Theft of Operational Information
									System Firmware		

Table 2. Precursor Analysis Report Quantitative Summary

Precursor Analysis Report Quantitative Summary	Totals
MITRE ATT&CK® for ICS Techniques	39
Technique Observables	743
Precursor Techniques	31
Precursor Technique Observables	574
Highly Perceivable Precursor Technique Observable	64

3. OBSERVABLE AND TECHNIQUE ANALYSIS

The following analysis may assist organizations in identifying malicious cyber activity earlier and more effectively. The following techniques and observables were compiled from publicly available sources and correlated with expert analysis.

3.1. REMOTE SERVICES (T0886) FOR INITIAL ACCESS

Adversaries using the INCONTROLLER framework may use the Remote Services technique for initial access to a victim's OT environment. Specifically, OMSHELL is capable of the remote services technique via the Telnet protocol to gain remote access to vulnerable Omron assets.⁷ However, it is unclear whether an adversary would require prior access to that environment or if they would be able to brute force weak credentials to internet-exposed assets to gain access.

IT Cybersecurity and OT Cybersecurity personnel may be able to observe anomalous Telnet traffic over TCP Port 23 in their environments due to OMSHELL's capability to access Omron assets from either a remote server or an infected IT environment.

A total of nine observables were identified with the use of the Remote Services technique (T0886). This technique is important for investigation because improperly configured OT networks can grant an adversary access without them having to compromise and pivot from an organization's IT environment. This technique appears early in the timeline and responding to it will likely limit the avenues an adversary can use to gain access into the OT environment. Terminating the chain of techniques at this point would halt future events by an adversary employing OMSHELL.

Of the nine observables associated with this technique, one is assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 24 artifacts could be generated by the Remote Services technique
Technique Observers^b	IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

^b Observer titles are adapted from the Job Role Groupings listed in [the SANS ICS Job Role to Competency Level Poster](#). CyOTE products utilize these job categories rather than organizational titles to both support comprehensive analysis and preserve anonymity within the victim organization. A complete list of potential observers can be found in [Appendix C](#).

3.2. USER EXECUTION TECHNIQUE (T0863) FOR EXECUTION

The adversary responsible for developing the INCONTROLLER framework would likely leverage the User Execution technique (T0863) to deploy ICECORE into the target's IT environment. ICECORE is likely a generic remote access tool developed by the adversary to bolster initial IT reconnaissance to eventually pivot into the OT environment. ICECORE uses Windows Management Instrumentation (WMI) to gain insight into the victim's enterprise environment. WMI also has a command-line equivalent, referred to as WMIC. Both sophisticated and unsophisticated groups of adversaries regularly abuse WMI due to the wide array of malicious activities that WMI can facilitate, such as data collection on infected hosts and remote code execution without interacting with the infected device's hard drive.⁸ WMI also relies on Server Message Block (SMB) and Remote Procedure Call Service (RPCS) over Port 135 for remote access, presenting a potential observable opportunity for defenders.

IT Cybersecurity and IT Staff personnel may be able to observe anomalous network traffic to adversary infrastructure over Ports 80 and 443 generated by ICECORE due to its theorized use as a general backdoor. Additionally, IT Cybersecurity and IT staff may observe anomalous WMI queries over TCP Ports 135 within the IT environment.

A total of 10 observables were identified with the use of the User Execution technique (T0863). This technique is important for investigation because unsuspecting end users often are the first observers in the chain of either a ransomware attack or an OT cybersecurity incident. This technique appears early in the timeline and responding to it will likely limit the avenues an adversary can use to gain access into the OT environment. Terminating the chain of techniques at this point may halt future events.

Of the 10 observables associated with this technique, two are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 23 artifacts could be generated by the User Execution technique
Technique Observers	IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.3. SCRIPTING TECHNIQUE (T0853) FOR EXECUTION

An adversary employing the INCONTROLLER framework would use the Scripting technique (T0853) to employ various tools within the framework. TAGRUN (MOUSEHOLE) uses an OPC-UA Python library to help identify OPC UA clients and servers in a victim environment, which then is compiled into a logfile and exfiltrated to an adversary-controlled C2 server.⁹ Additionally, Dragos states that TAGRUN, OMSHELL, CODECALL were written in Python as well.¹⁰

IT Cybersecurity, OT Staff, and OT Cybersecurity personnel may be able to observe anomalous Python scripts in their environment, although due to the many places an adversary can hide these files until needed for execution, it is unlikely that these anomalous python scripts would be observed

A total of 14 observables were identified with the use of the Scripting technique (T0853). This technique is important for investigation because the INCONTROLLER framework is largely Python-based and designed to be executed via the command line. This technique appears throughout the timeline, and responding to it will likely halt adversary activity in the OT environment. Terminating the chain of techniques at this point may halt future events.

Of the 14 observables associated with this technique, four are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Scripting technique
Technique Observers	IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.4. COMMAND-LINE EXECUTION TECHNIQUE (T0807) FOR EXECUTION

An adversary employing the INCONTROLLER framework almost certainly would use the CommandLine Execution technique (T0807) to execute various tools contained within the framework. TAGRUN that targets OPC UA clients and servers, OMSHELL that targets Omron PLCs, and CODECALL that targets Schneider Electric PLCs are designed to be executed via a Command Line Interface.¹¹ Finally, LAZYCARGO (CVE-2020-15368) requires a command line to exploit the ASRock driver vulnerability on a Windows host.¹²

IT Cybersecurity and OT Cybersecurity personnel may be able to observe anomalous instances of command prompts in their environment.

A total of 34 observables were identified with the use of the Command Line Interface technique (T0807). This technique is important for investigation because it indicates the adversary is actively attempting to manually execute a malicious payload in the victim's environment. This technique appears throughout the timeline and responding to it may halt further adversary activity within the victim's environment, as well as prevent execution of malware that may result in disruption or destruction. Terminating the chain of techniques at this point may mitigate both further malicious activity within a victim's environment and limit operational damage.

Of the 34 observables associated with this technique, five are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 25 artifacts could be generated by the Command Line Execution technique
Technique Observers	IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.5. NATIVE API TECHNIQUE (T0834) FOR EXECUTION

An adversary deploying the INCONTROLLER framework may use the Native API technique (T0834) to facilitate malicious activity in a victim's environment by leveraging native Windows tools. ICECORE leverages WMI to facilitate discovery, execution, and evasion from defenders. WMI is a native Windows tool that adversaries often use for reconnaissance, remote code execution, lateral movement, and persistence without the need to write any files to the disk. For remote access, WMI relies on Server Message Block (SMB) and Remote Procedure Call Service (RPCS) over Port 135 for remote access, presenting a potential observable opportunity for defenders.

IT Staff and IT Cybersecurity personnel may be able to observe adversary activity in WMI Event Viewer, although this is unlikely due to how often WMI is used for legitimate activity on the Windows OS.

A total of three observables were identified with the use of the Native API technique (T0834). This technique is important for investigation, because in this instance WMI is a very effective tool for an adversary due to the breadth of its capability. This technique appears early in the timeline and responding to it will delay or deny adversary reconnaissance, privilege escalation, and pivoting into the OT environment. Terminating the chain of techniques at this point would likely halt further adversary activity such as reconnaissance in a victim's IT environment.

Of the three observables associated with this technique, none are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 25 artifacts could be generated by the Native API technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.6. CHANGE OPERATING MODE (T0858) FOR EXECUTION

Adversaries using the INCONTROLLER framework may use the Change Operating mode technique for execution. OMSHELL can change operating modes of Omron PLCs for execution, and an adversary can manipulate exposed HTTP endpoints of Omron PLCs to change the operating mode of one or multiple targeted PLCs from “Program” to “Run.”¹³

OT Staff and OT Cybersecurity personnel may be able to observe anomalous HTTP traffic related to Omron’s proprietary FINS protocol. Additionally, OT Staff and OT Cybersecurity may also be able to observe anomalous operating mode changes on Omron PLCs.

A total of 10 observables were identified with the use of the Change Operating Mode technique (T0890). This technique is important for investigation because it demonstrates the adversary has lower-level access to crucial systems and assets in the OT environment. This technique may appear throughout the timeline and responding to it will prevent adversary privilege escalation. Terminating the chain of techniques at this point may halt some future events.

Of the 10 observables associated with this technique, none are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 24 artifacts could be generated by the Change Operating Mode technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.7. MODIFY PROGRAM TECHNIQUE (T0889) FOR PERSISTENCE

An adversary deploying the INCONTROLLER malware framework may use the Modify Program technique to bolster persistence in a victim's environment. The OMSHELL module's functionality would allow it to modify Omron devices to allow an adversary to upload a file to remotely connect and issue commands via HTTP traffic. CODECALL is also able to remove, modify, and replace logic on targeted Schneider Electric devices.¹⁴

OT Staff and OT Cybersecurity personnel may be able to observe anomalous HTTP or FINS network traffic related to modifying Omron PLCs.

A total of 33 observables were identified with the use of the Modify Program technique (T0889). This technique is important for investigation because an adversary is actively attempting to maintain remote access within the victim's OT environment to facilitate additional malicious activity. This technique appears somewhat early in the timeline and responding to it will prevent an adversary from maintaining continued access to the OT environment. Terminating the chain of techniques at this point would limit adversary access to OT assets.

Of the 33 observables associated with this technique, none are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 3 artifacts could be generated by the Modify Program technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.8. VALID ACCOUNTS TECHNIQUE (T0859) FOR PERSISTENCE

Numerous modules within the INCONTROLLER framework can use the Valid Accounts technique to bolster persistence within a victim's environment. Both OMSHELL and CODECALL are capable of using harvested credentials to interact with OT assets such as Engineering Workstations or HMIs.

OT Cybersecurity personnel may be able to observe anomalous logins at unusual hours from either the enterprise environment or external adversary command and control infrastructure.

A total of 19 observables were identified with the use of the Valid Accounts technique (T0859). This technique is important for investigation because malicious use of valid accounts is a stealthy way for an adversary to maintain persistence in a victim's environment. This technique may appear throughout the timeline, and responding to it may limit an adversary's access to the OT environment and OT assets. Terminating the chain of techniques at this point would limit the access and actions an adversary can take in the OT environment.

Of the 19 observables associated with this technique, four are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Valid Accounts technique
Technique Observers	OT Cybersecurity
Resources	Technique Detection References

3.9. EXPLOITATION FOR PRIVILEGE ESCALATION TECHNIQUE (T0890) FOR PRIVELEGE ESCALATION

The adversary behind the creation of the INCONTROLLER framework would likely use Exploitation for Privilege Escalation within the victim's environment by leveraging CVE-2020-15368, a vulnerability in an ASRock driver, allowing kernel-level execution of malicious code on an infected host.¹⁵

IT Cybersecurity and OT Cybersecurity personnel are not likely to observe the adversary leveraging this kernel-level vulnerability in an environment.

A total of four observables were identified with the use of the Exploitation for Privilege Escalation technique (T0890). This technique is important for investigation because it results in the adversary gaining root level access to crucial systems and assets in either the IT or OT environment. This technique appears somewhat early in the timeline, and responding to it will delay adversary privilege escalation. Terminating the chain of techniques at this point would limit an adversary's ability to conduct further malicious activity requiring higher privileges on a device or network.

Of the four observables associated with this technique, none are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Exploitation for Privilege Escalation technique
Technique Observers	IT Staff, OT Cybersecurity
Resources	Technique Detection References

3.10. CHANGE OPERATING MODE TECHNIQUE (T0858) FOR EVASION

Adversaries deploying the INCONTROLLER malware framework can employ the Change Operating Mode technique: OMSHELL can change the operation mode on Omron controllers, including the Monitor, Run, and Program modes.¹⁶

OT Staff and OT Cybersecurity personnel may be able to observe anomalous network traffic related to an adversary changing the operation mode of targeted PLCs, as well as anomalous system behavior on targeted Omron PLCs.

A total of 37 observables were identified with the use of the Change Operating Mode technique (T0858). This technique is important for investigation because an adversary may change operating modes to cover up any prior malicious activities such as reconnaissance, and responding to it may limit an adversary's reach in a targeted OT environment. Terminating the chain of techniques at this point may halt future adversary activity.

Of the 35 observables associated with this technique, none are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 24 artifacts could be generated by the Change Operating Mode technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.11. ROOTKIT TECHNIQUE (T0851) FOR EVASION

Adversaries employing the INCONTROLLER framework may use the LAZYCARGO module's Rootkit capability to bolster evasion. The adversary can install the signed AsrDrv103.sys driver that can act as a rootkit in a victim's IT or OT environment.¹⁷ This driver is vulnerable to CVE-2020-15368, allowing an adversary to execute code within the Windows kernel after installation of the vulnerable driver. Some ASRock motherboards may be used in certain OT assets such as HMIs and Engineering workstations, although this CVE may be used in both enterprise or operations environments.

IT Cybersecurity and OT Cybersecurity personnel may be able to observe anomalous program installation of either the AsrDrv103.sys file or additional payloads.

A total of four observables were identified with the use of the Rootkit technique (T0851). This technique is important for investigation as rootkits are one of the stealthiest methods an adversary can use to conduct malicious activity in a victim's environment. Responding to it will likely limit the ways an adversary can evade defender detection. Terminating the chain of techniques at this point would limit an adversary's options for persisting in a victim's environment.

Of the four observables associated with this technique, none are assessed to be highly perceivable. They are listed in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 23 artifacts could be generated by the Rootkit technique
Technique Observers	IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.12. REMOTE SYSTEM DISCOVERY TECHNIQUE (T0846) FOR DISCOVERY

Numerous modules within the INCONTROLLER framework can conduct remote system discovery within a victim's environment. ICECORE can use WMI for reconnaissance in the enterprise environment, likely identifying infrastructure suitable for pivoting into the OT environment or acting as a proxy node to external command and control infrastructure. TAGRUN uses OPC-UA to identify all OPC-UA configured clients and servers within a victim's environment via an ICMP Ping sweep.¹⁸ OMSHELL primarily uses FINS over User Datagram Protocol (UDP) to identify any Omron devices before conducting further probing via HTTP. CODECALL likely uses an initial batch of Address Resolution Protocol (ARP) calls to identify any networked devices and then use a UDP multicast over Port 27127 to identify any networked Schneider Electric devices.¹⁹

IT Staff, IT Cybersecurity, OT Staff, and OT Cybersecurity personnel may be able to observe anomalous network traffic related to WMI queries from ICECORE, anomalous OPC-UA traffic, anomalous UDP broadcasts, or anomalous ARP calls in the OT environment.

A total of 33 observables were identified with the use of Remote System Discovery technique (T0846). This technique is important for investigation because this technique indicates an adversary is actively reconnoitering, and responding to it will prevent an adversary from gaining insight into the architecture of a victim's OT environment. Terminating the chain of techniques at this point would limit the scope that an adversary can gain about an environment.

Of the 33 observables associated with this technique, one is assessed to be highly perceivable. It is italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 43 artifacts could be generated by the Remote System Discovery technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.13. REMOTE SYSTEM INFORMATION DISCOVERY TECHNIQUE (T0888) FOR DISCOVERY

Several modules within the INCONTROLLER malware framework are capable of the Remote System Information Discovery technique (T0888). ICECORE's WMI scanning can likely enumerate detailed information about hosts within the enterprise environment. TAGRUN's OPC UA scanning also can query specific information about OPC-configured assets to bolster adversary reconnaissance. OMSHELL primarily uses HTTP GET and POST requests to interact with Omron devices previously identified by FINS over UDP requests. CISA also notes that OMSHELL can parse HTTP traffic from OMRON devices to gain further insight into the victim's environment. CODECALL will send Modbus and CODESYS traffic to identify any networked assets that respond to the Modbus and CODESYS requests, demonstrating that CODECALL not only affects Schneider Electric devices, but those that use Modbus or CODESYS.

IT Cybersecurity, OT Staff, and OT Cybersecurity personnel may be able to observe anomalous network traffic linked to various remote system information discovery activity as well as the presence of anomalous files related to adversary activity.

A total of 35 observables were identified with the use of the Remote System Information Discovery technique (T0888). This technique is important for investigation because it allows an adversary to gain detailed information of critical OT assets. This technique appears throughout the timeline and responding to it may limit the amount of detail an adversary can learn about assets in a victim's environment. Terminating the chain of techniques at this point would limit the insight an adversary can gain into specific assets within an environment.

Of the 35 observables associated with this technique, two are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 8 artifacts could be generated by the Remote System Information Discovery technique
Technique Observers	IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.14. NETWORK SNIFFING TECHNIQUE (T0842) FOR DISCOVERY

The OMSHELL module within the INCONTROLLER framework is capable of the Network Sniffing technique (T0842). CISA notes that OMSHELL can parse HTTP response headers from Omron devices, allowing an adversary to glean detailed information about Omron assets in the victim's environment.²⁰ OMSHELL is also able to perform general network traffic captures as well as retrieve the MAC addresses of Omron devices.

OT Staff and OT Cybersecurity personnel may be able to observe anomalous files on infected hosts. Although INCONTROLLER was mostly composed of Python scripts when it was discovered, this could change in a real-world deployment.

A total of 14 observables were identified with the use of the Network Sniffing technique (T0842). This technique is important for investigation because an adversary is attempting to gain insights into the OT environment by snooping on plaintext network traffic. This technique appears in the middle of timeline and responding to it will limit the options an adversary can take for reconnaissance. Terminating the chain of techniques at this point may halt further adversary activity.

Of the 14 observables associated with this technique, one is assessed to be highly perceivable. It is italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 11 artifacts could be generated by the Network Sniffing technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.15. VALID ACCOUNTS TECHNIQUE (T0859) FOR LATERAL MOVEMENT

Numerous modules within the INCONTROLLER framework can use the Valid Accounts technique (T0859) for lateral movement. The TAGRUN module can brute force logins into OPC UA servers using a hardcoded list of common passwords.²¹ CODECALL can also brute force logins to Schneider PLCs over UDP Port 1740.²²

OT Staff and OT Cybersecurity personnel may be able to observe logons with valid accounts at anomalous hours, as well as repeated failed logons. Observers may also be able to detect anomalous UDP traffic over Port 1740.

A total of 11 observables were identified with the use of the Valid Accounts technique (T0859). This technique is important for investigation because an adversary will often use default or valid credentials to facilitate lateral movement into higher-privileged areas in the network. This technique can appear throughout the timeline and responding to it may halt further activity. The chain of techniques at this point would limit an adversary's capability to move laterally throughout the victim's OT environment.

Of the 11 observables associated with this technique, four are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Valid Accounts technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.16. DEFAULT CREDENTIALS TECHNIQUE (T0812) FOR LATERAL MOVEMENT

Several modules within INCONTROLLER leverage the Default Credentials technique (T0812) to facilitate lateral movement. OMSHELL and CODECALL can leverage hardcoded default credentials that Omron and Schneider Electric Devices come with from the factory. CODECALL may use CODESYS over UDP Port 1740 to brute force logins with default credentials or common passwords via a dictionary file.²³ Default credentials are often as simple as “password,” “admin,” or “root”.

OT Staff and OT Cybersecurity personnel may be able to observe anomalous failed logins or brute forcing of Omron and Schneider Electric device interfaces. OT Staff and OT Cybersecurity may also observe anomalous failed login attempts against devices that use CODESYS due to CODECALL’s ability to interact with CODESYS-compatible devices.

A total of 19 observables were identified with the use of the Default Credentials technique (T0812). This technique is important for investigation because adversaries can easily use hardcoded factory credentials that were not changed to gain further access into a victim’s OT environment. This technique can appear throughout the timeline and responding to it may halt all future events. Terminating the chain of techniques at this point may halt an adversary’s options to conduct additional malicious activity in a victim’s environment.

Of the 19 observables associated with this technique, three are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of XX artifacts could be generated by the Default Credentials technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.17. PROGRAM DOWNLOAD TECHNIQUE (T0843) FOR LATERAL MOVEMENT

The OMSHELL module in the INCONTROLLER framework is capable of the Program Download technique (T0843). OMSHELL can load an attacker-customized configuration file to conduct malicious activity on infected Omron devices. CODECALL is also able to download adversary-customized files over CODESYS to a targeted PLC.²⁴

OT Staff and OT Cybersecurity personnel may be able to observe anomalous network traffic associated with downloading an external program. OT Staff and OT Cybersecurity may also observe anomalous files on Schneider Electric and Omron devices, or engineering workstations.

A total of 23 observables were identified with the use of the Program Download technique (T0843). This technique is important for investigation as the adversary can load malicious software that can be used to cause a disruptive or destructive effect on OT assets. Responding to it may halt all future events for Omron and Schneider devices, as well as those that use the Modbus or CODESYS protocols. Terminating the chain of techniques at this point would limit any operational damage or loss of productivity or revenue.

Of the 23 observables associated with this technique, none are assessed to be highly perceivable. They are listed in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 19 artifacts could be generated by the Program Download technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.18. LATERAL TOOL TRANSFER TECHNIQUE (T0867) FOR LATERAL MOVEMENT

An adversary using the INCONTROLLER framework has multiple ways to use the lateral tool transfer technique within a victim's environment. ICECORE may be used with other tools such as TAGRUN, CODECALL, LAZYCARGO, and OMSHELL from an adversary-controlled command and control server once an adversary has established persistence in the IT environment and access into the OT environment.

IT Cybersecurity and OT Cybersecurity personnel may be able to observe anomalous network traffic to and from C2 servers, as well as anomalous network traffic between the IT and OT environments.

A total of five observables were identified with the use of the Lateral Tool Transfer technique (T0867). This technique is important for investigation because the adversary is attempting to move OT-specific offensive tools into the victim's environment. This technique appears somewhat early in the timeline and responding to it may halt disruption or damage to OT assets. Terminating the chain of techniques at this point would limit additional malicious adversary activity in the OT environment as well as future disruption or damage to OT equipment.

Of the five observables associated with this technique, two are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 22 artifacts could be generated by the Lateral Tool Transfer technique
Technique Observers	IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.19. REMOTE SERVICES TECHNIQUE (T0886) FOR LATERAL MOVEMENT

The OMSHELL module within the INCONTROLLER framework can abuse remote services to facilitate lateral movement. OMSHELL uses the telnet and FINS protocols to gain remote access to vulnerable OT assets such as HMIs and engineering workstations.

IT Cybersecurity and OT Cybersecurity personnel may be able to observe anomalous telnet or FINS traffic in their environments due OMSHELL's capability to access OT assets from either a remote server or an infected IT environment.

A total of 13 observables were identified with the use of the Remote Services technique (T0886). This technique is important because an adversary is attempting to move through a victim's environment to conduct additional malicious activity. This technique may appear throughout the timeline and responding to it would limit an adversary's options for access into the victim's environment. Terminating the chain of techniques at this point may halt future events.

Of the 13 observables associated with this technique, two assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 24 artifacts could be generated by the Remote Services technique
Technique Observers	IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.20. POINT AND TAG IDENTIFICATION TECHNIQUE (T0861) FOR COLLECTION

The TAGRUN module within the INCONTROLLER framework is capable of the Point and Tag Identification technique (T0861). OPC uses labels known as points and tags that reference various aspects of either an OPC client or OPC server. Points might refer to values, inputs, outputs, or process-specific assets. Tags typically are labels given to OPC assets for operator convenience. An adversary would be interested in mapping OPC points and tags to assist in reconnaissance of a target OT environment or to actively interact with OPC assets.

OT Cybersecurity personnel may be able to observe anomalous TCP/IP traffic associated with malicious OPC UA traffic related to the Point and Tag Identification technique.

A total of eight observables were identified with the use of the Point and Tag Identification technique (T0861). This technique is important for investigation as an adversary using this technique is conducting reconnaissance and attempting to learn more about the architecture of the OT environment. Terminating the chain of techniques at this point may halt future events, as an adversary would have limited options to conduct comprehensive reconnaissance of an OT environment.

Of the eight observables associated with this technique, four are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 53 artifacts could be generated by the Point and Tag Identification technique
Technique Observers	OT Cybersecurity
Resources	Technique Detection References

3.21. MONITOR PROCESS STATE TECHNIQUE (T0801) FOR COLLECTION

The TAGRUN module within the INCONTROLLER framework is capable of the Monitor Process State technique (T0801). TAGRUN can query the state of OPC clients and servers within the OT environment over 4840 via read requests and read responses. The adversary would then likely export this data to a file that would be sent off to a C2 server to facilitate payload development and testing. OMSHELL is also capable of monitoring process states of Omron devices using a shell over Telnet and Omron's proprietary FINS protocol.²⁵

OT Staff and OT Cybersecurity personnel likely would not be able to observe anomalous OPC UA traffic over port 4840. However, the prior anomalous ICMP Ping requests may also act as a triggering event before the Read Requests and Read Responses.

A total of 22 observables were identified with the use of the Monitor Process State technique (T0801). This technique is important for investigation because an adversary is attempting to collect operational information on the victim's OT environment. This technique appears throughout the timeline and responding to it may limit the methods of reconnaissance an adversary may undertake. Terminating the chain of techniques at this point may prevent the adversary from exfiltrating operational information.

Of the 22 observables associated with this technique, five are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Monitor Process Technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.22. PROGRAM UPLOAD TECHNIQUE (T0845) FOR COLLECTION

OMSHELL is cable of the Program Upload technique to support collection in a victim's OT environment. An adversary employing OMSHELL would be able to harvest Omron PLC configuration files and load customized adversary-crafted configuration files. CODECALL may also be able to upload information deemed critical by an adversary via CODESYS or Modbus.

OT Cybersecurity personnel may be able to observe anomalous configuration files on engineering workstations or the anomalous network traffic associated with file uploads.

A total of 35 observables were identified with the use of the Program Upload technique (T00845). This technique is important for investigation because an adversary is attempting to gain crucial information on OT assets within a targeted environment, and responding to it may halt an adversary's reconnaissance activities. Terminating the chain of techniques at this point may halt some future events.

Of the 35 observables associated with this technique, one is assessed to be highly perceivable. It is italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 14 artifacts could be generated by the Program Upload technique
Technique Observers	OT Cybersecurity
Resources	Technique Detection References

3.23. DETECT OPERATING MODE TECHNIQUE (T0868) FOR COLLECTION

OMSHELL is capable of the Detect Operating Mode technique (T0868). OMSHELL likely uses Omron's FINS protocol to determine the operating mode of Omron PLCs within a targeted OT environment.²⁶

OT Staff and OT Cybersecurity personnel may be able to observe anomalous FINS network traffic.

A total of 22 observables were identified with the use of the Detect Operating Mode technique (T0868). This technique is important for investigation and an adversary is actively mapping out the victim's OT infrastructure and understanding configurations for future payload developments. This technique appears in the middle of the timeline and responding to it may halt future events. Terminating the chain of techniques at this point would likely limit the amount of useful information an adversary could gather.

Of the 22 observables associated with this technique, none are assessed to be highly perceivable. The observables are listed in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 8 artifacts could be generated by the Detect Operating Mode technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.24. STANDARD APPLICATION LAYER PROTOCOL TECHNIQUE (T0869) FOR COMMAND AND CONTROL

The INCONTROLLER framework makes use of several standard application layer protocols in its various modules. ICECORE may use HTTP for command-and-control traffic between an infected host and the adversary. Malware with similar features of ICECORE often abuse HTTP and HTTPS due to the ability to blend into legitimate network traffic. TAGRUN uses the Internet Control Message Protocol (ICMP) and uses Ping to identify networked devices.²⁷ OMSHELL may use both UDP and HTTPS for reconnaissance and command traffic between target devices and the threat actor. CODECALL also uses Address Resolution Protocol (ARP) calls to identify any networked devices before sending out Modbus and CODESYS traffic.²⁸

IT Cybersecurity, OT Staff, OT Cybersecurity personnel may be able to observe anomalous network traffic associated with various protocols such as anomalous ICMP Ping requests and responses, ARP calls and responses, and UDP broadcast traffic in the OT environment.

A total of 51 observables were identified with the use of the Standard Application Layer protocol technique (T0869). This technique is important for investigation because network access is key to an adversary conducting malicious activity in a target environment. Responding to it will typically halt all future events. Terminating the chain of techniques at this point would limit adversarial further activity and prevent any disruption or damage to the OT environment.

Of the 51 observables associated with this technique, six are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Standard Application Layer protocol technique
Technique Observers	IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.25. COMMONLY USED PORT TECHNIQUE (T0885) FOR COMMAND AND CONTROL

Numerous modules within the INCONTROLLER framework use the Commonly Used Port technique (T0885) to support C2 activities with an adversary. CyOTE analysts assess that ICECORE may be like other Windows malware and likely uses Ports 80 and 443 for C2 traffic, as well as Port 135 for WMI discovery activities. TAGRUN likely uses Port 7 for its ICMP Ping discovery capability, as well as TCP Port 4840 to scan for OPC UA servers, but this port can easily be changed by an adversary.²⁹ As a part of the proprietary FINS protocol, OMSHELL may use Port 23 for Telnet and Port 80 for FINS, which is based on HTTP. OMSHELL may also use Port 9600 over both TCP and UDP. CODECALL will initially send out UDP broadcasts over Port 27127 in the OT environment; however, since this is the standard scanning port, observers should use discretion when analyzing anomalous scanning on 27127.³⁰ CODECALL can also scan for Schneider Electric devices over UDP on Port 1740-1743, TCP Port 1105, and TCP Port 11740.³¹ CODECALL also scans for Modbus devices on Port 502.³²

IT Cybersecurity, OT Staff, and OT Cybersecurity personnel may be able to observe anomalous network traffic over the aforementioned network ports.

A total of 51 observables were identified with the use of the Commonly Used Port technique (T0885). This technique is important for investigation as it presents an opportunity for defenders to identify an adversary's command and control traffic. This technique appears throughout the timeline and responding to it may halt future malicious activity. Terminating the chain of techniques at this point would limit adversary activity and potentially halt some future events.

Of the 51 observables associated with this technique, six are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 5 artifacts could be generated by the Commonly Used Port technique
Technique Observers	IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.26. DENIAL OF SERVICE TECHNIQUE (T0814) FOR INHIBIT RESPONSE FUNCTION

The CODECALL module within the INCONTROLLER framework is capable of the Denial of Service technique. CODECALL can carry out a Distributed Denial of Service (DDoS) attack by disabling CODESYS connections to targeted PLCs and sending custom packets to crash any targeted PLCs.³³ OMSHELL can also conduct DDoS attacks to terminate any active connections to Omron PLCs.

OT Staff and OT Cybersecurity personnel may be able to observe either anomalous network connection errors or network connection failures.

A total of 11 observables were identified with the use of the Denial of Service technique (T0814). This technique is important for investigation because an adversary is attempting to disrupt a properly functioning device, which may result in operational disruption or, in some cases, physical damage. This technique appears late in the timeline and responding to it likely will not halt all future events. Terminating the chain of techniques at this point may limit some operational disruption.

Of the 11 observables associated with this technique, two are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 14 artifacts could be generated by the Denial of Service technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.27. DATA DESTRUCTION TECHNIQUE (T0809) FOR INHIBIT RESPONSE FUNCTION

The CODECALL module within the INCONTROLLER framework is capable of the Data Destruction technique. CODECALL can not only execute files on Schneider, CODESYS, and Modbus-enabled devices, but can delete files, as well.

OT Staff and OT Cybersecurity personnel may be able to observe anomalous network traffic related to file deletion, as well as critical files missing from affected devices.

A total of 17 observables were identified with the use of the Data Destruction technique (T0809). This technique is important for investigation because an adversary is attempting to inhibit properly functioning equipment within the OT environment and responding to it may prevent further disruption or damage. Terminating the chain of techniques at this point may potentially limit operational disruption or physical damage.

Of the 17 observables associated with this technique, none are assessed to be highly perceivable. The observables are listed in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 27 artifacts could be generated by the Data Destruction technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.28. SERVICE STOP TECHNIQUE (T0801) FOR INHIBIT RESPONSE FUNCTION

The OMSHELL module within the INCONTROLLER framework is capable of the Service Stop technique. OMSHELL may be used to kill arbitrary processes running on Omron devices to disrupt any physical processes controlled by Omron PLCs.³⁴

OT Staff and OT Cybersecurity personnel may be able to observe anomalous process disruption, as well as the network traffic associated with this disruption.

A total of eight observables were identified with the use of the Service Stop technique (T0801). This technique is important for investigation because the adversary is actively attempting to stop critical functions related to service or productivity. This technique appears late in the timeline, and responding to it may halt some future events. Terminating the chain of techniques at this point would limit loss of productivity and revenue, physical process disruption, or damage.

Of the eight observables associated with this technique, one is assessed to be highly perceivable. It is italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 13 artifacts could be generated by the Service Stop technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.29. MODIFY PARAMETER TECHNIQUE (T0836) FOR IMPAIR PROCESS CONTROL

Two modules within INCONTROLLER are capable of the Modify Parameter technique. CODECALL can read and write custom commands over Modbus to affected devices, as well as devices that rely on CODESYS to modify device parameters.^{35,36} OMSHELL can modify the parameters of affected Omron devices that control physical processes such as motor speed or functions that detect overspeed conditions.

OT Staff and OT Cybersecurity personnel may be able to observe anomalous behavior from Omron devices, as well as those that use Modbus and CODESYS.

A total of 20 observables were identified with the use of the Modify Parameter technique (T0836). This technique is important for investigation because an adversary is actively tampering with parameters crucial for maintaining process integrity within the OT environment. This technique appears late in the timeline and responding to it may prevent further disruption within the OT environment. Terminating the chain of techniques at this point may limit operational disruption or damage.

Of the 20 observables associated with this technique, three are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 18 artifacts could be generated by the Modify Parameter technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.30. UNAUTHORIZED COMMAND MESSAGE TECHNIQUE (T0855) FOR IMPAIR PROCESS CONTROL

Two modules within the INCONTROLLER framework can use the Unauthorized Command Message technique (T0855). TAGRUN can not only read OPC UA values, but also write them. However, CyOTE analysts assess that TAGRUN is tool for reconnaissance, and is unlikely to be used in the final stage of an OT cyberattack. The CODECALL module can send Modbus and CODESYS packets to Schneider Electric devices, as well as other brands of devices that can accept Modbus traffic.³⁷

OT Staff and OT Cybersecurity personnel would likely not be able to observe malicious OPC UA values being written to devices or malicious Modbus and CODESYS packets in time to prevent operational disruption.

A total of 15 observables were identified with the use of the Unauthorized Command Message technique (T0855). This technique is important for investigation because an adversary is attempting to manipulate OT assets. This technique appears late in the timeline and responding to it is unlikely to halt future events. Terminating the chain of techniques at this point in the timeline is unlikely to prevent disruption or destruction.

Of the 15 observables associated with this technique, two are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Unauthorized Command Message technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.31. DEVICE SHUTDOWN OR RESTART (T0816) FOR IMPAIR PROCESS CONTROL

CODECALL is capable of the Device Shutdown or Restart technique (T0816). CODECALL can crash CODESYS devices and conduct a “packet of death” attack which may result in the device needing to be restarted.^{38,39}

OT Staff and OT Cybersecurity personnel likely would not be able to observe anomalous reboots or malicious Modbus and CODESYS packets in time to prevent operational disruption.

A total of 10 observables were identified with the use of the Unauthorized Command Message technique (T0816). This technique is important for investigation because an adversary is attempting to interfere with normal processes in the OT environment. This technique appears late in the timeline and responding to it is not likely to halt future events. Terminating the chain of techniques at this point in the timeline is not likely to prevent disruption or destruction.

Of the 10 observables associated with this technique, three are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 17 artifacts could be generated by the Unauthorized Command Message technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.32. THEFT OF OPERATIONAL INFORMATION TECHNIQUE (T0882) FOR IMPACT

Multiple modules within INCONTROLLER are capable of the Theft of Operational Information technique (T0882). TAGRUN can identify a victim's OT architecture by using OPC UA to map out OPC clients and servers throughout an OT environment. OMSHELL can capture and record network traffic related to Omron PLCs, parse network traffic for specific attributes such as MAC addresses, and export this to an adversary. CODECALL is also able to download configuration files from Schneider Electric PLCs as well as those that use CODESYS, likely to assist an adversary in developing customized payloads.

IT Cybersecurity, OT Staff, and OT Cybersecurity personnel are not likely to observe the precursor reconnaissance activity as anomalous files present on hosts that would be sent off to an adversary. However, an adversary is likely to delete any files related to malicious scanning or network traffic captures which may present an observable opportunity for defenders

A total of 52 observables were identified with the use of the Theft of Operational Information technique (T0882). This technique is important for investigation because the exfiltration of operational information is almost certain to bolster payload development by an adversary. This technique appears in the middle of the timeline and responding to it will limit how much an adversary can learn about a victim's OT environment. Terminating the chain of techniques at this point may halt some future events, however an adversary would likely continue to have access as well as the necessary information to develop customized payloads.

Of the 52 observables associated with this technique, seven are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 4 artifacts could be generated by the Theft of Operational Information technique
Technique Observers	IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.33. DENIAL OF CONTROL TECHNIQUE (T0813) FOR IMPACT

The CODECALL module within the INCONTROLLER framework is capable of the Denial of Control technique (T0813). CODECALL can sever CODESYS connections and disconnect active sessions with affected PLCs. CODECALL can crash logic on the controller, preventing any communications to the PLC unless the power is cycled.

OT Staff and OT Cybersecurity personnel may observe unresponsive OT systems.

A total of 11 observables were identified with the use of the Denial of Control technique (T0813). This technique is important for investigation because the adversary is actively conducting malicious activity within the OT environment and preventing authorized personnel and safety systems from preventing disruption or physical damage. This technique appears late in the timeline and responding to it is not likely to halt future events. Terminating the chain of techniques at this point is not likely to limit any operational disruption or physical damage.

Of the 11 observables associated with this technique, two are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 8 artifacts could be generated by the Denial of Control technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.34. DENIAL OF VIEW TECHNIQUE (T0815) FOR IMPACT

The CODECALL module within the INC ONTROLLER framework is capable of the Denial of View technique (T0815). CODECALL can crash PLCs several different ways, and communications between engineering workstations and the crashed PLCs may cause a temporary denial of view scenario. A power cycle and configuration recovery are necessary to restore monitoring and communication to any affected PLCs.⁴⁰

OT Staff and OT Cybersecurity personnel may observe anomalous instances where OT systems are unresponsive to communications.

A total of 11 observables were identified with the use of the Denial of View technique (T0815). This technique is important for investigation because operators and safety systems lose visibility into OT assets responsible for physical processes. This technique appears late in the timeline and responding to it may hasten operators regaining view and control of OT assets. Terminating the chain of techniques at this point is not likely to prevent operational disruption but may prevent physical damage.

Of the 11 observables associated with this technique, two are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 4 artifacts could be generated by the Denial of View technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.35. LOSS OF AVAILABILITY TECHNIQUE (T0826) FOR IMPACT

The CODECALL module within the INCONTROLLER framework is capable of the Loss of Availability technique (T0826). Due to CODECALL's ability to sever communications and cause a temporary denial of view scenario, this could simultaneously result in a loss of availability until remedial and recovery efforts are taken.

OT Staff, OT Cybersecurity personnel may observe unresponsive OT systems as well as anomalous behavior in physical processes.

A total of 11 observables were identified with the use of the Loss of Availability technique (T0826). This technique is important for investigation because an adversary is attempting to disrupt OT systems to cause operational disruption or, in some cases, physical damage. This technique appears late in the timeline and responding to it would not have any effect. Terminating the chain of techniques at this point is not likely to limit operational disruption or physical damage.

Of the 11 observables associated with this technique, two are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 8 artifacts could be generated by the Loss of Availability technique
Technique Observers	OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.36. LOSS OF PRODUCTIVITY AND REVENUE (T0828) FOR IMPACT

If deployed in a comprehensive manner, INCONTROLLER is capable of the Loss of Productivity and Revenue technique (T0828). Several of INCONTROLLER's modules can cause either operational disruption or physical destruction, leading to unplanned downtime.

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may observe operational disruption or physical destruction that would impact productivity and revenue flow.

A total of two observables were identified with the use of the Loss of Productivity and Revenue technique (T0828). This technique is important for investigation because it involves potential damage to not only systems, but business losses as well. This technique appears at the end of the attack and responding to it will not have any preventative effect. Terminating the chain of techniques at this point would not have any effect.

Of the two observables associated with this technique, both are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 5 artifacts could be generated by the Loss of Productivity technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management
Resources	Technique Detection References

3.37. MANIPULATION OF CONTROL TECHNIQUE (T0831) FOR IMPACT

Several modules within the INCONTROLLER framework are capable of the Manipulation of Control technique (T0831). TAGRUN can write values to OPC clients and servers, allowing an adversary to manipulate any OT assets that are reliant on OPC UA. CODECALL is also able to manipulate Schneider Electric devices, as well as those that use Modbus and CODESYS. Finally, OMSHELL can manipulate both Omron PLCs as well as some Omron servo drives.⁴¹

OT Staff and OT Cybersecurity may observe anomalous behavior from OPC, Omron, and Schneider Electric devices, as well as devices that use CODESYS and Modbus.

A total of 22 observables were identified with the use of the Manipulation of Control technique (T0828). This technique is important for investigation because an adversary is actively manipulating OT assets to cause either operational disruption or physical damage, and responding to it is not likely to prevent damage or disruption. Terminating the chain of techniques at this point would not have a major effect.

Of the 22 observables associated with this technique, 10 are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Manipulation of Control technique
Technique Observers	OT Staff, OT Cybersecurity

3.38. DAMAGE TO PROPERTY TECHNIQUE (T0879) FOR IMPACT

The INCONTROLLER framework can cause damage to property if deployed by a capable adversary. This may include damage to physical equipment that is connected to Schneider Electric and Omron PLCs, as well as equipment reliant on OPC UA, Modbus, and CODESYS protocols.

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may observe physical damage to equipment caused by the full deployment of the INCONTROLLER framework.

A total of 1 observable was identified with the use of the Damage to Property technique (T0979). This technique is important for investigation because damage to property may increase the physical danger to OT Staff and other personnel or cause a loss of productivity and revenue. This technique appears last in the timeline and responding to it likely will not have an effect regardless of termination of the chain of events.

The one observable associated with this technique is assessed to be highly perceivable, it is italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 18 artifacts could be generated by the Damage to Property technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management

3.39. LOSS OF SAFETY TECHNIQUE (T0837) FOR IMPACT

OMSHELL may be capable of the Loss of Safety technique (T0837). In a fictitious scenario, OMSHELL may be used to disable PLCs responsible for maintaining a safe OT environment, such as Omron's NX SL3300, as well as to reprogram other ICS and OT assets to cause operational disruption or destruction.

OT Staff, OT Cybersecurity, Engineering, and Support Staff personnel may have been able to observe unresponsive systems or sustained loss of control in the OT environment.

A total of four observables were identified with the use of the Loss of Safety technique (T0837). This technique is important for investigation because an adversary disabling safety systems poses an immediate threat to personnel and property. This technique appears late in the timeline and responding to it in a timely manner may prevent operational disruption or damage. Terminating the chain of techniques at this point may halt future events if responded to in a timely manner.

Of the four observables associated with this technique, three are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 3 artifacts could be generated by the Loss of Safety technique
Technique Observers	OT Staff, OT Cybersecurity, Engineering, Support Staff

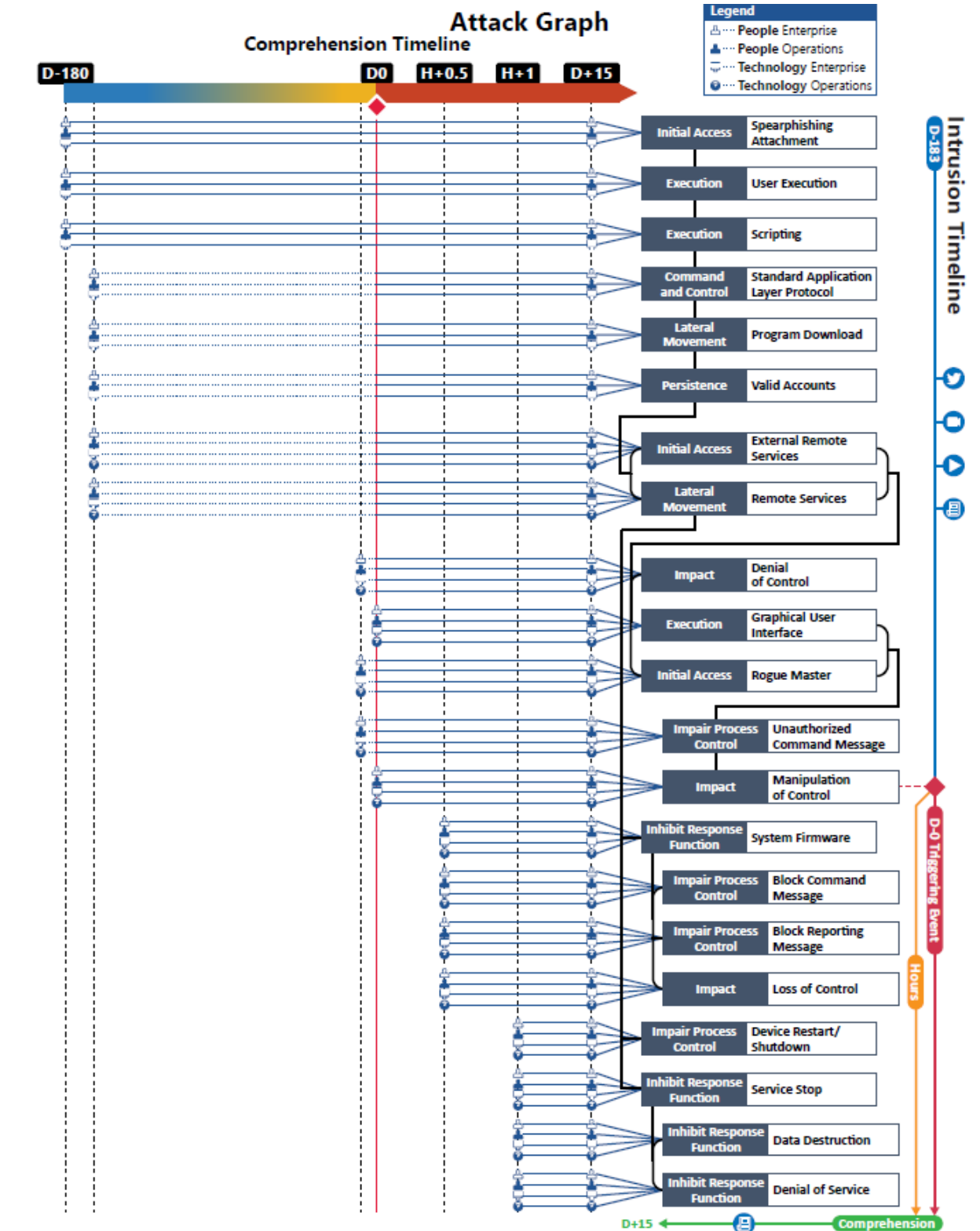


Figure 2. Attack Graph

APPENDIX A: OBSERVABLES LIBRARY

NOTE: Highly perceivable observables are italicized and marked †

Observables Associated with the Remote Services Technique (T0886)	
Observable 1	Presence of Anomalous Binary on Host
Observable 2	Execution of Anomalous Binary on Host
Observable 3	Presence of Anomalous Background Process
Observable 4	Presence of Anomalous Background Process: Telnet Daemon
Observable 5	Anomalous Network Traffic
Observable 6	Anomalous Network Traffic: From remote server to local client
Observable 7†	<i>Anomalous Network Traffic: From remote server to local client: Over TCP Port 23 Telnet</i>
Observable 8	Anomalous Command Line
Observable 9	Anomalous Command Line: Execution of anomalous commands
Observable 10	Observable details.

Observables Associated with the User Execution Technique (T0863)	
Observable 1	Anomalous Activity on Host
Observable 2†	<i>Anomalous Activity on Host: Registry Modification (Event ID 4657)</i>
Observable 3†	<i>Anomalous Network Traffic: Inbound: HTTP: Over TCP Port 80: HTTP GET Request</i>
Observable 4	Anomalous Network Traffic: Inbound: HTTPS: Over TCP Port 443: HTTP GET Request
Observable 5	Anomalous Network Traffic: Outbound: HTTPS: Over TCP Port 443: HTTP GET Request
Observable 6	Anomalous Network Traffic: Outbound: HTTPS: Over TCP Port 80: HTTP GET Request
Observable 7	Presence of Anomalous Binary on Host
Observable 8	Anomalous Registry Changes
Observable 9	Anomalous Network Traffic: Inbound: WMI: Over TCP Port 135
Observable 10	Anomalous Network Traffic: Outbound: HTTPS: Over TCP Port 135

Observables Associated with the Scripting Technique (T0853)	
Observable 1 †	Presence of Anomalous Files on Local Host
Observable 2	Presence of Anomalous Files on Local Host: Python scripts
Observable 3	Presence of Anomalous Files on Local Host: Linux ELF File
Observable 4	Anomalous Command Line Interface [E1]
Observable 5	Anomalous Command line Interface: Python Scripts

Observables Associated with the Scripting Technique (T0853)	
Observable 6	Anomalous Command line Interface: ELF file
Observable 7	Execution of Anomalous Executable
Observable 8	Execution of Anomalous Executable: Mismatch of Parent and Child Process (Process ID - PID)
Observable 9	Installation of Anomalous Software Package
Observable 10	Installation of Anomalous Software Package: Python Interpreter
Observable 11†	<i>Creation of Windows Event ID: 4688 - New Process has Been Created</i>
Observable 12†	<i>Anomalous Creation of Registry Key</i>
Observable 13†	<i>Windows Event ID 4657 - Modification of Registry Key</i>
Observable 14†	<i>Anomalous Modification of Registry Key</i>

Observables Associated with the Command Line Execution Technique (T0807)	
Observable 1	Anomalous Command Line
Observable 2	Anomalous Command Line: Python Execution
Observable 3	Anomalous Command line: WMI Execution
Observable 4	Anomalous Command line: WMIC Execution
Observable 5	Presence of Anomalous Command Line Interface [E1]
Observable 6	Presence Anomalous Command line Interface: Python Scripts
Observable 7	Execution of Anomalous Executable
Observable 8	Execution of Anomalous Executable: Mismatch of Parent and Child Process (Process ID - PID)
Observable 9	Installation of Anomalous Software Package
Observable 10	Installation of Anomalous Software Package: Python Interpreter
Observable 11†	<i>Creation of Windows Event ID: 4688 - New Process has Been Created</i>
Observable 12†	<i>Anomalous Creation of Registry Key</i>
Observable 13†	<i>Windows Event ID 4657 - Modification of Registry Key</i>
Observable 14†	<i>Anomalous Modification of Registry Key</i>
Observable 15	Presence of Anomalous Binary on Host
Observable 16	Execution of Anomalous Binary on Host
Observable 17	Presence of Anomalous Background Process
Observable 18	Presence of Anomalous Background Process: Telnet Daemon
Observable 19	Anomalous Network Traffic
Observable 20	Anomalous Network Traffic: From remote server to local client
Observable 21†	<i>Anomalous Network Traffic: From remote server to local client: Over TCP Port 23 Telnet</i>

Observables Associated with the Command Line Execution Technique (T0807)	
Observable 22	Anomalous Network Traffic: From remote server to local client: Over TCP Port 80 HTTP
Observable 23	Anomalous Network Traffic: From remote server to local client: Over TCP Port 80 HTTP: Factory Interface Network Service (FINS) Protocol
Observable 24	Anomalous Command Line: Execution of anomalous commands
Observable 25	Anomalous Command Line: Execution of Python Scripts
Observable 26	Presence of Anomalous Executable on Host
Observable 27	Presence of Anomalous Files on Local Host
Observable 28	Presence of Anomalous Files on Local Host: Python scripts
Observable 29	Presence of Anomalous Files on Local Host: Linux ELF File
Observable 30	Anomalous Command Line Interface
Observable 31	Anomalous Command Line Interface: Python Scripts [E2]
Observable 32	Presence Anomalous Command line Interface: ELF file
Observable 33	Process Creation (Sysmon Event ID 1) [E3]
Observable 34	File Creation (Sysmon Event ID 11) [E3]

Observables Associated with the Native API Technique (T0834)	
Observable 1	Anomalous Activity on Host
Observable 2	Anomalous Activity on Host: WMI
Observable 3	Anomalous Activity on Host: WMIC

Observables Associated with the Change Operating Mode Technique (T0858)	
Observable 1 †	Anomalous Command Line
Observable 2	Anomalous Command Line: Execution of Python Scripts
Observable 3	Anomalous Command Line: Execution of anomalous commands
Observable 4	Anomalous Network Traffic
Observable 5	Anomalous Network Traffic: From Internal Host to Local Control Device
Observable 6	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 7	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 8	Anomalous Network Traffic: From Remote Host to Local Control Device
Observable 9	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)

Observables Associated with the Change Operating Mode Technique (T0858)

Observable 10	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
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Observables Associated with the Modify Program Technique (T0889)

Observable 1 †	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Internal Host to Local Control Device
Observable 3	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 4	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 5	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Anomalous packet content
Observable 6	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Anomalous packet content: Filename source Name
Observable 7	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Anomalous packet content: Filename Dest Name
Observable 8	Anomalous Network Traffic: From Remote Host to Local Control Device
Observable 9	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 10	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Anomalous packet content
Observable 12	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Anomalous packet content: Filename source Name
Observable 13	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Anomalous packet content: Filename Dest Name
Observable 14	Anomalous Command Line
Observable 15	Anomalous Command Line: Omron command 0x2203 (single file write)
Observable 16	Anomalous Command Line: Omron command 0x2211 (File Memory Write)
Observable 17	Anomalous Command Line: Omron command 0x2207 (File Copy)
Observable 18	Anomalous Network Traffic: From Local Host to Control Device
Observable 19	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 20	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502

Observables Associated with the Modify Program Technique (T0889)	
Observable 21	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 22	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 23	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 24	Anomalous Command Line: Execution of anomalous commands
Observable 25	Presence of Anomalous Files on Local Host
Observable 26	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host
Observable 27	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File
Observable 28	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cnf file
Observable 29	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .conf file
Observable 30	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cfg file
Observable 31	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cf file
Observable 32	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .ini file
Observable 33	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .json file

Observables Associated with the Valid Accounts Technique (T0859)	
Observable 1	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From remote server to local client
Observable 3	Anomalous Network Traffic: From remote server to local client: Over TCP Port 23 Telnet
Observable 4	Anomalous Command Line
Observable 5	Anomalous Command Line: Anomalous Failed Login Attempts
Observable 6	Anomalous Command Line: Anomalous Timing of Login with Valid Credentials
Observable 7	Anomalous Command Line: Anomalous Timing of Login with Valid Certificate
Observable 8	Anomalous Command Line: Anomalous Timing of Login via Valid Password
Observable 9	Anomalous Network Traffic: From Local Host to Control Device
Observable 10	Anomalous Network Traffic: From Local Host to Control Device: UDP
Observable 11	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1740
Observable 12	Anomalous Command Line on Local Host: Operational Technology (OT) Host

Observables Associated with the Valid Accounts Technique (T0859)	
Observable 13†	<i>Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Failed Login Attempts</i>
Observable 14	Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Timing of Login with Valid Credentials
Observable 15	Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Timing of Login with Valid Certificate
Observable 16	Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Timing of Login via Valid Password
Observable 17	Anomalous Host Activity: Successful Logon From External Host: An Account was Successfully Logged On Windows Event ID (4624)
Observable 18†	<i>Anomalous Host Activity: UnSuccessful Logon From External Host: An Account Failed to Log on Windows Event ID (4625)</i>

Observables Associated with the Exploitation for Privilege Escalation Technique (T0890)	
Observable 1	Anomalous Command Prompt
Observable 2	Anomalous Command Prompt: c:\asrock\work\asrocksdk_v0.0.69\asrrw\src\driver\src\objfre_win7_amd64\amd64\AsrDrv103.pdb
Observable 3	Presence of Anomalous Files on Host
Observable 4	Presence of Anomalous Files on Host: AsrDrv103.pdb

Observables Associated with the Change Operating Mode Technique (T0850)	
Observable 1	Anomalous Command Line
Observable 2	Anomalous Command Line: Execution of Python Scripts
Observable 3	Anomalous Command Line: Execution of anomalous commands
Observable 4	Anomalous Command Line: Execution of anomalous commands: rsync command
Observable 5	Anomalous Command Line: Execution of anomalous commands: dd command
Observable 6	Anomalous Command Line: Execution of anomalous commands: tar command
Observable 7	Anomalous Command Line: Execution of anomalous commands: gzip command
Observable 8	Anomalous Network Traffic
Observable 9	Anomalous Network Traffic: From Internal Host to Local Control Device
Observable 10	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 11	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol

Observables Associated with the Change Operating Mode Technique (T0850)	
Observable 12	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 HTTP: Factory Interface Network Service (FINS) Protocol: Filename source Name
Observable 13	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename Dest Name
Observable 14	Anomalous Network Traffic: From Remote Host to Local Control Device
Observable 15	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 16	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 17	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename source Name
Observable 18	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename Dest Name
Observable 19	Presence of Anomalous Files on Host
Observable 20	Presence of Anomalous Files on Host: .dd
Observable 21	Presence of Anomalous Files on Host: .tar
Observable 22	Presence of Anomalous Files on Host: .gz
Observable 23	Creation of Anomalous Files on Host
Observable 24	Creation of Anomalous Files on Host: .dd
Observable 25	Creation of Anomalous Files on Host: .tar
Observable 26	Creation of Anomalous Files on Host: .gz
Observable 27	Anomalous Command Line on Control Device
Observable 28	Anomalous Command Line on Control Device: Omron command 0x2203 (Single File Write)
Observable 29	Anomalous Command Line on Control Device: Omron command 0x2211 (File Memory Write)
Observable 30	Anomalous Command Line on Control Device: Omron command 0x2207 (File Copy)
Observable 31	Anomalous System Behavior on Control Device
Observable 32	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC)
Observable 33	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device
Observable 34	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device: Change in Operating Mode

Observables Associated with the Change Operating Mode Technique (T0850)

Observable 35	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device: Change in Operating Mode: Monitor Mode to Another Mode
Observable 36	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device: Change in Operating Mode: Run Mode to Another Mode
Observable 37	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device: Change in Operating Mode: Program Mode to Another Mode

Observables Associated with the Rootkit Technique (T0851)

Observable 1	Anomalous Command Prompt
Observable 2	Anomalous Command Prompt: c:\\asrock\\work\\asrocksdk_v0.0.69\\asrrw\\src\\driver\\src\\objfre_win7_amd64\\amd64\\AsrDrv103.pdb
Observable 3	Presence of Anomalous File on Host
Observable 4	Presence of Anomalous File on Host: AsrDrv103.pdb

Observables Associated with the Remote System Discovery Technique (T0846)

Observable 1	Anomalous Activity on Host
Observable 2	Anomalous Activity on Host: WMI
Observable 3	Anomalous Activity on Host: WMIC
Observable 4	Anomalous Network Traffic
Observable 5	Anomalous Network Traffic: From Client to Server: ICMP: Echo (Ping) Request Type 8
Observable 6	Anomalous Network Traffic: From Server to Client: ICMP: Echo (Ping) Reply Type Zero
Observable 7	Anomalous Network Traffic: OPC UA
Observable 8	Anomalous Network Traffic: OPC UA: From Client to Server: Hello Message
Observable 9	Anomalous Network Traffic: OPC UA: From Server to Client: Acknowledge Message
Observable 10	Anomalous Network Traffic: OPC UA: From Client to Server: Read Request
Observable 11	Anomalous Network Traffic OPC UA: From Server to Client: Read Response
Observable 12	Anomalous Network Traffic: Outbound: From OPC Client to Server: Port 4840
Observable 13	Anomalous Network Traffic: Inbound: From OPC Server to Client: Port 4840
Observable 14	Presence of Anomalous Files on Local Host
Observable 15	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host

Observables Associated with the Remote System Discovery Technique (T0846)	
Observable 16	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Script File
Observable 17	Anomalous Network Traffic
Observable 18	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device
Observable 19	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 20	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 21	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Recursive GET and POST Requests
Observable 22	Anomalous Command Line
Observable 23	Anomalous Command Line: Execution of anomalous commands
Observable 24	Anomalous Command Line: Execution of anomalous commands: tcpdump command
Observable 25	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Packet Capture File (.pcap)
Observable 26	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Packet Capture File Next Generation (.pcapng)
Observable 27	Anomalous Network Traffic: From Local Host to Control Device
Observable 28	Anomalous Network Traffic: From Local Host to Control Device: ARP call
Observable 29	Anomalous Network Traffic: From Control Device to Local Host
Observable 30	Anomalous Network Traffic: From Control Device to Local Host: ARP response
Observable 31	Anomalous Network Traffic: From Local Host to Control Device: UDP
Observable 32	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 27127
Observable 33	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 27127: Multicast

Observables Associated with the Remote System Information Discovery Technique (T0888)	
Observable 1	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Internet Control Message Protocol (ICMP) Echo Request Type 8
Observable 2	Anomalous Network Traffic: From Local Host to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) Hosts to Client Server with Python Executable : Internet Control Message Protocol (ICMP) Echo Reply Type 0

Observables Associated with the Remote System Information Discovery Technique (T0888)	
Observable 3	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Hello Message"
Observable 4	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Acknowledge Message"
Observable 5	Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml)
Observable 6	Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Configuration
Observable 7	Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Firmware Details
Observable 8	Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Application Details
Observable 9	Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Project Recipes
Observable 10	Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Control System States
Observable 11	Presence of Anomalous Files on Local Host
Observable 12	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host
Observable 13	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Python Script File
Observable 14	Anomalous Network Traffic
Observable 15	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device
Observable 16	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 17	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 18	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP):

Observables Associated with the Remote System Information Discovery Technique (T0888)	
	Factory Interface Network Service (FINS) Protocol: Recursive GET and POST Requests
Observable 19	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over UDP Port 9600 Omron SYSMAC Cx Product
Observable 20	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 9600 Omron SYSMAC Cx Product
Observable 21	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 23 Telnet
Observable 22	Anomalous Command Line on Local Host
Observable 23	Anomalous Command Line on Local Host: Operational Technology (OT) Host
Observable 24	Anomalous Command Line on Local Host: Operational Technology (OT) Host: Execution of Anomalous Commands
Observable 25 †	<i>Anomalous Command Line on Local Host: Operational Technology (OT) Host: Execution of Anomalous Commands: "tcpdump" Command</i>
Observable 26	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Packet Capture File (.pcap)
Observable 27	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Packet Capture File Next Generation (.pcapng)
Observable 28	Anomalous Network Traffic: From Local Host to Control Device
Observable 29	Anomalous Network Traffic: From Local Host to Control Device: UDP
Observable 30	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 27127
Observable 31	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 32	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 33	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 34	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 35	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS

Observables Associated with the Network Sniffing Technique (T0842)	
Observable 1	Presence of Anomalous Files on Local Host
Observable 2	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host
Observable 3	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Python Script File
Observable 4	Anomalous Network Traffic
Observable 5	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device

Observables Associated with the Network Sniffing Technique (T0842)	
Observable 6	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 7	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 8	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Recursive GET and POST Requests
Observable 9	Anomalous Command Line
Observable 10	Anomalous Command Line: Execution of anomalous commands
Observable 11 †	<i>Anomalous Command Line: Execution of anomalous commands: tcpdump command</i>
Observable 12	Presence of Anomalous Files on Local Host
Observable 13	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Packet Capture File (.pcap)
Observable 14	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Packet Capture File Next Generation (.pcapng)

Observables Associated with the Valid Accounts Technique (T0859)	
Observable 1 †	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Local Host to Control Device
Observable 3	Anomalous Network Traffic: From Local Host to Control Device: UDP
Observable 4	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1740
Observable 5	Anomalous Command Line on Local Host: Operational Technology (OT) Host
Observable 6 †	<i>Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Failed Login Attempts</i>
Observable 7	Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Timing of Login with Valid Credentials
Observable 8	Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Timing of Login with Valid Certificate
Observable 9	Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Timing of Login via Valid Password
Observable 10	Anomalous Host Activity: Successful Logon From External Host: An Account was Successfully Logged On Windows Event ID (4624)
Observable 11 †	<i>Anomalous Host Activity: UnSuccessful Logon From External Host: An Account Failed to Log on Windows Event ID (4625)</i>
Observable 12 †	<i>Anomalous Failed Login Attempts [E1, E3, E4]</i>
Observable 13	Anomalous Remote Logins to OPC Server [E1]
Observable 14 †	<i>Anomalous Failed Remote Logins to OPC Server [E1]</i>

Observables Associated with the Valid Accounts Technique (T0859)	
Observable 15	Anomalous timing of login with valid credentials [E2]
Observable 16	Anomalous timing of login via valid certificate [E2, E3]
Observable 17	Anomalous timing of login via valid password [E2]

Observables Associated with the Default Credentials Technique (T0812)	
Observable 1 †	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device
Observable 3	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 23 Telnet
Observable 4	Anomalous Command Line on Local Host
Observable 5	Anomalous Command Line on Local Host: Operational Technology (OT) Host
Observable 6 †	<i>Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Failed Login Attempts</i>
Observable 7	Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Timing of Login with Valid Credentials
Observable 8	Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Timing of Login with Valid Certificate
Observable 9	Anomalous Command Line on Local Host: Operational Technology (OT) Host: Anomalous Timing of Login via Valid Password
Observable 10	Anomalous Host Activity: Successful Logon From External Host: An Account was Successfully Logged On Windows Event ID (4624)
Observable 11 †	<i>Anomalous Host Activity: Unsuccessful Logon From External Host: An Account Failed to Log on Windows Event ID (4625)</i>
Observable 12	Anomalous Network Traffic
Observable 13	Anomalous Network Traffic: From Local Host to Control Device
Observable 14	Anomalous Network Traffic: From Local Host to Control Device: UDP
Observable 15	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1740
Observable 16	Presence of Anomalous Files on Local Host
Observable 17	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host
Observable 18	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Text File
Observable 19	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Text File: Dictionary File

Observables Associated with the Program Download Technique (T0843)	
Observable 1	Anomalous Network Traffic

Observables Associated with the Program Download Technique (T0843)	
Observable 2	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device
Observable 3	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 4	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 5	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Recursive GET and POST Requests
Observable 6	Presence of Anomalous Files on Local Host
Observable 7	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host
Observable 8	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File
Observable 9	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cnf file
Observable 10	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .conf file
Observable 11	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cfg file
Observable 12	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cf file
Observable 13	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .ini file
Observable 14	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .json file
Observable 15	Anomalous Network Traffic
Observable 16	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 17	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 18	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 19	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 20	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 21	Anomalous Command Line
Observable 22	Anomalous Command Line: Execution of anomalous commands
Observable 23	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Text Files

Observables Associated with the Lateral Tool Transfer Technique (T0867)

Observable 1	Anomalous Network Traffic
Observable 2 †	Anomalous Network Traffic: From Local Host to External IP: Over HyperText Transfer Protocol (HTTP) TCP Port 80
Observable 3 †	Anomalous Network Traffic: From External IP to Local Host: Over HyperText Transfer Protocol (HTTP) TCP Port 80
Observable 4	Anomalous Network Traffic: From Local Host to External IP: Over HyperText Transfer Secure Protocol (HTTPS) TCP Port 443
Observable 5	Anomalous Network Traffic: From External IP to Local Host: Over HyperText Transfer Protocol (HTTP) TCP Port 443

Observables Associated with the Remote Services Technique (T0886)

Observable 1	Presence of Anomalous Binary on Host
Observable 2	Execution of Anomalous Binary on Host
Observable 3	Presence of Anomalous Background Process
Observable 4	Presence of Anomalous Background Process: Telnet Daemon
Observable 5	Anomalous Network Traffic
Observable 6	Anomalous Network Traffic: From remote server to local client
Observable 7 †	<i>Anomalous Network Traffic: From remote server to local client: Over TCP Port 23 Telnet</i>
Observable 8	Anomalous Command Line
Observable 9	Anomalous Command Line: Execution of anomalous commands
Observable 10 †	<i>Anomalous Command Line: Anomalous Failed Login Attempts</i>
Observable 11	Anomalous Command Line: Anomalous Timing of Login with Valid Credentials
Observable 12	Anomalous Command Line: Anomalous Timing of Login with Valid Certificate
Observable 13	Anomalous Command Line: Anomalous Timing of Login via Valid Password

Observables Associated with the Point and Tag Identification Technique (T0861)

Observable 1	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Internet Control Message Protocol (ICMP) Echo Request Type 8
Observable 2	Anomalous Network Traffic: From Local Host to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) Hosts to Client Server with Python Executable: Internet Control Message Protocol (ICMP) Echo Reply Type 0
Observable 3	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Read Request"

Observables Associated with the Point and Tag Identification Technique (T0861)	
Observable 4	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Read Response"
Observable 5 †	<i>Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml)</i>
Observable 6 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml)</i>
Observable 7 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Configuration</i>
Observable 8 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Control System States</i>

Observables Associated with the Monitor Process Technique (T0801)	
Observable 1 †	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Read Request"
Observable 2	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Read Response"
Observable 3 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Control System States</i>
Observable 4	Anomalous Command Line
Observable 5	Anomalous Command Line: Execution of anomalous commands
Observable 6	Anomalous Network Traffic
Observable 7	Anomalous Network Traffic: From Internal Host to Local Control Device
Observable 8	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 9	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 10	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 HTTP: Factory Interface Network Service (FINS) Protocol: Filename source Name

Observables Associated with the Monitor Process Technique (T0801)	
Observable 11	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename Dest Name
Observable 12	Anomalous Network Traffic: From Remote Host to Local Control Device
Observable 13	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 14	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 15	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename source Name
Observable 16	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename Dest Name
Observable 17	Anomalous Command Line on Control Device
Observable 18	Anomalous System Behavior on Control Device
Observable 19 †	<i>Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC)</i>
Observable 20 †	<i>Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device</i>
Observable 21 †	<i>Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device: Change in Operating Mode</i>
Observable 22 †	<i>Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device: Change in Operating Mode: Current Mode to Monitor Mode</i>

Observables Associated with the Program Upload Technique (T0845)	
Observable 1 †	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 3	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 4	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 5	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 6	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 7	Anomalous Command Line
Observable 8	Anomalous Command Line: Execution of anomalous commands
Observable 9	Presence of Anomalous Files on Local Host

Observables Associated with the Program Upload Technique (T0845)	
Observable 10	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host
Observable 11	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File
Observable 13	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cnf file
Observable 14	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .conf file
Observable 15	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cfg file
Observable 16	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cf file
Observable 17	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .ini file
Observable 18	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .json file
Observable 19	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Text Files
Observable 20	Presence of Anomalous Files on Host
Observable 21	Presence of Anomalous Files on Local Host: Python scripts
Observable 22	Anomalous Command Line: Execution of Python Scripts
Observable 23	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device
Observable 24	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 23 Telnet
Observable 25	Anomalous Command Line: Execution of anomalous commands: Omron command 0x2207 (File Copy)
Observable 26	Anomalous Command Line: Execution of anomalous commands: Omron command 0x2203 (single file write)
Observable 27	Anomalous Command Line: Execution of anomalous commands: Omron command 0x2211 (File Memory Write)
Observable 28	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 29	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 30	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 HTTP: Factory Interface Network Service (FINS) Protocol: Filename source Name
Observable 31	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename Dest Name

Observables Associated with the Program Upload Technique (T0845)	
Observable 32	Anomalous Network Traffic: From Remote Host to Local Control Device
Observable 33	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 34	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 35	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename source Name

Observables Associated with the Detect Operating Mode Technique (T0868)	
Observable 1	Anomalous Command Line
Observable 2	Anomalous Command Line: Execution of anomalous commands
Observable 3	Anomalous Network Traffic
Observable 4	Anomalous Network Traffic: From Internal Host to Local Control Device
Observable 5	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 6	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 7	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 HTTP: Factory Interface Network Service (FINS) Protocol: Filename source Name
Observable 8	Anomalous Network Traffic: From Internal Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename Dest Name
Observable 9	Anomalous Network Traffic: From Remote Host to Local Control Device
Observable 10	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 11	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 13	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename source Name
Observable 14	Anomalous Network Traffic: From Remote Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol: Filename Dest Name
Observable 15	Anomalous Command Line on Control Device
Observable 16	Anomalous System Behavior on Control Device
Observable 17	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC)

Observables Associated with the Detect Operating Mode Technique (T0868)

Observable 18	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device
Observable 19	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device: Detect Operating Mode
Observable 20	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device: Provides Operating Mode
Observable 21	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device: Provides Operating Mode: Run Mode
Observable 22	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Omron Device: Provides Operating Mode: Monitor Mode

Observables Associated with the Standard Application Layer Protocol Technique (T0869)

Observable 1	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Local Host to External IP: Over HyperText Transfer Protocol (HTTP) TCP Port 80
Observable 3	Anomalous Network Traffic: From External IP to Local Host: Over HyperText Transfer Protocol (HTTP) TCP Port 80
Observable 4	Anomalous Network Traffic: From Local Host to External IP: Over HyperText Transfer Secure Protocol (HTTPS) TCP Port 443
Observable 5	Anomalous Network Traffic: From External IP to Local Host: Over HyperText Transfer Secure Protocol (HTTPS) TCP Port 443
Observable 6	Anomalous Network Traffic: From Local Host to External IP: WMI: Port 135
Observable 7	Anomalous Network Traffic: From External IP to Local Host: WMI: Port 135
Observable 8	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Internet Control Message Protocol (ICMP) Echo Request Type 8
Observable 9	Anomalous Network Traffic: From Local Host to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) Hosts to Client Server with Python Executable: Internet Control Message Protocol (ICMP) Echo Reply Type 0
Observable 10	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Hello Message"
Observable 11	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Acknowledge Message"
Observable 13	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture

Observables Associated with the Standard Application Layer Protocol Technique (T0869)	
	(UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Read Request"
Observable 14	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Read Response"
Observable 15	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Write Request"
Observable 16	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Write Response"
Observable 17	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Call Method Request"
Observable 18	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Call Method Response"
Observable 19	Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml)
Observable 20	Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml)
Observable 21	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device
Observable 22	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 23 Telnet
Observable 23	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 24	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 25 †	<i>Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 23 Telnet</i>
Observable 26	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 27	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 28	Anomalous Network Traffic: From Local Host to Control Device: TCP

Observables Associated with the Standard Application Layer Protocol Technique (T0869)	
Observable 29	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 30	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 31	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 32	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 33	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS: Anomalous Malformed Packet
Observable 34	Anomalous Network Traffic: From Local Host to Control Device: UDP
Observable 35	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 27127
Observable 36	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1740
Observable 37	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1741
Observable 38	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1742
Observable 39	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1743
Observable 40	Anomalous Network Traffic: From Control Device to Local Host: TCP
Observable 41	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 502
Observable 42	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 502: MODBUS
Observable 43	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 11740
Observable 44	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 11740: CODESYS
Observable 45	Anomalous Network Traffic: From Control Device to Local Host: UDP
Observable 46	Anomalous Network Traffic: From Control Device to Local Host: UDP: Port 27127
Observable 47	Anomalous Network Traffic: From Control Device to Local Host: UDP: Port 1740
Observable 48	Anomalous Network Traffic: From Control Device to Local Host: UDP: Port 1741
Observable 49	Anomalous Network Traffic: From Control Device to Local Host: UDP: Port 1742
Observable 50	Anomalous Network Traffic: From Control Device to Local Host: UDP: Port 1743
Observable 51	Anomalous Command Line
Observable 52	Anomalous Command Line: Execution of anomalous commands

Observables Associated with the Commonly Used Port Technique (T0885)	
Observable 1	Anomalous Network Traffic
Observable 2 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over HyperText Transfer Protocol (HTTP) TCP Port 80</i>

Observables Associated with the Commonly Used Port Technique (T0885)	
Observable 3 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Hypertext Transfer Protocol (HTTP) TCP Port 80</i>
Observable 4	Anomalous Network Traffic: From Local Host to External IP: Over HyperText Transfer Secure Protocol (HTTPS) TCP Port 443
Observable 5	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Internet Control Message Protocol (ICMP) Echo Request Type 8
Observable 6	Anomalous Network Traffic: From Local Host to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) Hosts to Client Server with Python Executable: Internet Control Message Protocol (ICMP) Echo Reply Type 0
Observable 7	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Hello Message"
Observable 8	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Acknowledge Message"
Observable 9	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Read Request"
Observable 10	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Read Response"
Observable 11	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Write Request"
Observable 13	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Write Response"
Observable 14	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Call Method Request"
Observable 15	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Call Method Response"

Observables Associated with the Commonly Used Port Technique (T0885)	
Observable 16 †	<i>Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml)</i>
Observable 17 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml)</i>
Observable 18	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device
Observable 19 †	<i>Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 23 Telnet</i>
Observable 20	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 21	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 22	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over UDP Port 9600 Omron SYSMAC Cx Product
Observable 23	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 9600 Omron SYSMAC Cx Product
Observable 24 †	<i>Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 23 Telnet</i>
Observable 25	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 26	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 27	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over UDP Port 9600 Omron SYSMAC Cx Product
Observable 28	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 9600 Omron SYSMAC Cx Product
Observable 29	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 30	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 31	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 32	
Observable 33	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 34	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 35	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS: Anomalous Malformed Packet
Observable 36	Anomalous Network Traffic: From Local Host to Control Device: UDP

Observables Associated with the Commonly Used Port Technique (T0885)	
Observable 37	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 27127
Observable 38	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1740
Observable 39	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1741
Observable 40	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1742
Observable 41	Anomalous Network Traffic: From Local Host to Control Device: UDP: Port 1743
Observable 42	Anomalous Network Traffic: From Control Device to Local Host: TCP
Observable 43	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 502
Observable 44	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 502: MODBUS
Observable 45	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 11740
Observable 46	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 11740: CODESYS
Observable 47	Anomalous Network Traffic: From Control Device to Local Host: UDP
Observable 48	Anomalous Network Traffic: From Control Device to Local Host: UDP: Port 27127
Observable 49	Anomalous Network Traffic: From Control Device to Local Host: UDP: Port 1740
Observable 50	Anomalous Network Traffic: From Control Device to Local Host: UDP: Port 1741
Observable 51	Anomalous Network Traffic: From Control Device to Local Host: UDP: Port 1742
Observable 52	Anomalous Network Traffic: From Control Device to Local Host: UDP: Port 1743

Observables Associated with the Denial of Service Technique (T0814)	
Observable 1	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 3	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 4	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 5	Anomalous Command Line
Observable 6	Anomalous Command Line: Execution of anomalous commands
Observable 7	Anomalous Host Activity
Observable 8 †	<i>Anomalous Host Activity: Legitimate Process Anomally Stopped</i>
Observable 9 †	<i>Anomalous Host Activity: OT Systems Unresponsive</i>
Observable 10	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)

Observables Associated with the Denial of Service Technique (T0814)

Observable 11	Anomalous Network Traffic: From Internal Operational Technology (OT) Host to Local Control Device: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
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Observables Associated with the Data Destruction Technique (T0809)

Observable 1	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 3	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 4	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 5	Anomalous Command Line
Observable 6	Anomalous Command Line: Execution of anomalous commands
Observable 7	Anomalous Host Activity
Observable 8	Anomalous Host Activity: OT Environment
Observable 9	Anomalous Host Activity: OT Environment: Anomalous Deletion of Files
Observable 10	Anomalous Host Activity: OT Environment: Anomalous Deletion of Files: Configuration File
Observable 11	Anomalous Host Activity: OT Environment: Anomalous Deletion of Files: Configuration File: .cnf file
Observable 12	Anomalous Host Activity: OT Environment: Anomalous Deletion of Files: Configuration File: .conf file
Observable 13	Anomalous Host Activity: OT Environment: Anomalous Deletion of Files: Configuration File: .cfg file
Observable 14	Anomalous Host Activity: OT Environment: Anomalous Deletion of Files: Configuration File: .cf file
Observable 15	Anomalous Host Activity: OT Environment: Anomalous Deletion of Files: Configuration File: .ini file
Observable 16	Anomalous Host Activity: OT Environment: Anomalous Deletion of Files: Configuration File: .json file
Observable 17	Anomalous Host Activity: OT Environment: Anomalous Deletion of Files: Text Files

Observables Associated with the Service Stop Technique (T0801)

Observable 1	Anomalous Host Activity
Observable 2 †	<i>Anomalous Host Activity: Legitimate Process Anomally Stopped</i>
Observable 3	Presence of Anomalous Files on Local Host
Observable 4	Presence of Anomalous Files on Local Host: Python scripts
Observable 5	Anomalous Network Traffic

Observables Associated with the Service Stop Technique (T0801)

Observable 6	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host
Observable 7	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 8	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol

Observables Associated with the Modify Parameter Technique (T0836)

Observable 1	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 3	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 4	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 5	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 6	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 7	Anomalous Command Line
Observable 8	Anomalous Command Line: Execution of anomalous commands
Observable 9 †	<i>Anomalous System Behavior on Control Device</i>
Observable 10 †	<i>Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC)</i>
Observable 11	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Anomalous Allocation of Memory Space
Observable 12	Presence of Anomalous Files on Host
Observable 13	Presence of Anomalous Files on Local Host: Python scripts
Observable 14	Anomalous Command Line: Execution of Python Scripts
Observable 15	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host
Observable 16	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 17	Anomalous Network Traffic: From Local Control Device to Internal Operational Technology (OT) Host: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 18	Anomalous Command Line: Execution of anomalous commands: 0x220F (File Memory Index Read)
Observable 19	Anomalous Command Line: Execution of anomalous commands: 0x2211 (File Memory Write)

Observables Associated with the Modify Parameter Technique (T0836)

Observable 20	Anomalous Command Line: Execution of anomalous commands: 0x0103 (Memory Area Fill)
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Observables Associated with the Unauthorized Command Message Technique (T0855)

Observable 1	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Write Request"
Observable 2	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Write Response"
Observable 3 †	<i>Anomalous Host Activity: Local Devices Connected to Client Server: Open Communication Platform (OPC) Unified Architecture (UA) Device Associated with Local Open Communication Platform (OPC) Unified Architecture (UA) Client Server: Client Requiring Repeated Authentication</i>
Observable 4	Anomalous Network Traffic
Observable 5	Anomalous Network Traffic: OPC UA
Observable 6	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 7	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 8	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 9	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 10	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 11	Anomalous Command Line
Observable 12	Anomalous Command Line: Execution of anomalous commands
Observable 13	Anomalous System Behavior on Control Device
Observable 14	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC)
Observable 15	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Anomalous Allocation of Memory Space

Observables Associated with the Device Shutdown or Restart Technique (T0816)

Observable 1	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 3	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740

Observables Associated with the Device Shutdown or Restart Technique (T0816)

Observable 4	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 5	Anomalous Command Line
Observable 6	Anomalous Command Line: Execution of anomalous commands
Observable 7	Anomalous Host Activity
Observable 8 †	<i>Anomalous Host Activity: Legitimate Process Anomally Stopped</i>
Observable 9 †	<i>Anomalous Host Activity: OT Systems Unresponsive</i>
Observable 10 †	<i>Anomalous Host Activity: Anomalous OT system Reboot</i>

Observables Associated with the Theft of Operational Information Technique (T0882)

Observable 1	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Internet Control Message Protocol (ICMP) Echo Request Type 8
Observable 2	Anomalous Network Traffic: From Local Host to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) Hosts to Client Server with Python Executable: Internet Control Message Protocol (ICMP) Echo Reply Type 0
Observable 3	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Hello Message"
Observable 4	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Acknowledge Message"
Observable 5	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Read Request"
Observable 6	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Read Response"
Observable 7	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Write Request"
Observable 8	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Write Response"

Observables Associated with the Theft of Operational Information Technique (T0882)	
Observable 9	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Call Method Request"
Observable 10	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Call Method Response"
Observable 11 †	<i>Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml)</i>
Observable 13 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml)</i>
Observable 14 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Configuration</i>
Observable 15 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Firmware Details</i>
Observable 16 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Application Details</i>
Observable 17 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Project Recipes</i>
Observable 18 †	<i>Presence of Sysmon Event ID 1 [E2]: Presence of Anomalous Application Log Files: Open Platform Communication (OPC) Unified Architecture (UA) Log Files: Extensible Markup Language (.xml): Containing Operational Information: Control System States</i>
Observable 19	Presence of Anomalous Files on Host
Observable 20	Presence of Anomalous Files on Local Host: Python scripts
Observable 21	Anomalous Command Line on Local Host
Observable 22	Presence of Anomalous Files on Local Host
Observable 23	Presence of Anomalous Files on Local Host: Packet Capture File (.pcap)
Observable 24	Presence of Anomalous Files on Local Host: Packet Capture File Next Generation (.pcapng)
Observable 25	Presence of Anomalous Files on Local Host: .dd
Observable 26	Presence of Anomalous Files on Local Host: .tar
Observable 27	Presence of Anomalous Files on Local Host: .gz

Observables Associated with the Theft of Operational Information Technique (T0882)	
Observable 28	Presence of Anomalous Files on Local Host: Configuration File: .cnf file
Observable 29	Presence of Anomalous Files on Local Host: Configuration File: .conf file
Observable 30	Presence of Anomalous Files on Local Host: Configuration File: .cfg file
Observable 31	Presence of Anomalous Files on Local Host: Configuration File: .cf file
Observable 32	Presence of Anomalous Files on Local Host: Configuration File: .ini file
Observable 33	Presence of Anomalous Files on Local Host: Configuration File: .json file
Observable 34	Anomalous Network Traffic
Observable 35	Anomalous Network Traffic: From Internal Host to Remote Host
Observable 36	Anomalous Network Traffic: From Internal Host to Remote Host: Over TCP Port 80 Hypertext Transfer Protocol (HTTP)
Observable 37	Anomalous Network Traffic: From Internal Host to Remote Host: Over TCP Port 80 Hypertext Transfer Protocol (HTTP): Factory Interface Network Service (FINS) Protocol
Observable 38	Anomalous Network Traffic: From Control Device to Local Host: TCP
Observable 39	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 502
Observable 40	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 502: MODBUS
Observable 41	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 11740
Observable 42	Anomalous Network Traffic: From Control Device to Local Host: TCP: Port 11740: CODESYS
Observable 43	Anomalous Command Line
Observable 44	Anomalous Command Line: Execution of anomalous commands
Observable 45	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host
Observable 46	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File
Observable 47	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cnf file
Observable 48	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .conf file
Observable 49	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cfg file
Observable 50	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .cf file
Observable 51	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .ini file
Observable 52	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Configuration File: .json file

Observables Associated with the Theft of Operational Information Technique (T0882)

Observable 53	Presence of Anomalous Files on Local Host: Operational Technology (OT) Network Host: Text Files
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Observables Associated with the Denial of Control Technique (T0813)

Observable 1 †	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 3	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 4	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 5	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 6	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 7	Anomalous Command Line
Observable 8	Anomalous Command Line: Execution of anomalous commands
Observable 9	Anomalous Host Activity
Observable 10 †	<i>Anomalous Host Activity: Legitimate Process Anomally Stopped</i>
Observable 11 †	<i>Anomalous Host Activity: OT Systems Unresponsive</i>

Observables Associated with the Denial of View Technique (T0815)

Observable 1	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 3	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 4	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 5	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 6	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 7	Anomalous Command Line
Observable 8	Anomalous Command Line: Execution of anomalous commands
Observable 9	Anomalous Host Activity
Observable 10 †	<i>Anomalous Host Activity: Legitimate Process Anomally Stopped</i>
Observable 11 †	<i>Anomalous Host Activity: OT Systems Unresponsive</i>

Observables Associated with the Loss of Availability Technique (T0826)	
Observable 1	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 3	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 4	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 5	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 6	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 7	Anomalous Command Line
Observable 8	Anomalous Command Line: Execution of anomalous commands
Observable 9	Anomalous Host Activity
Observable 10 †	<i>Anomalous Host Activity: Legitimate Process Anomally Stopped</i>
Observable 11 †	<i>Anomalous Host Activity: OT Systems Unresponsive</i>

Observables Associated with the Loss of Productivity and Revenue (T0828)	
Observable 1 †	<i>Anomalous Loss of Productivity</i>
Observable 2 †	<i>Anomalous Loss of Revenue</i>

Observables Associated with the Manipulation of Control Technique (T0831)	
Observable 1 †	Anomalous Network Traffic
Observable 2	Anomalous Network Traffic: OPC UA
Observable 3	Anomalous Network Traffic: From Local Host to Control Device: TCP
Observable 4	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502
Observable 5	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 502: MODBUS
Observable 6	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740
Observable 7	Anomalous Network Traffic: From Local Host to Control Device: TCP: Port 11740: CODESYS
Observable 8	Anomalous Command Line
Observable 9	Anomalous Command Line: Execution of anomalous commands
Observable 10 †	<i>Anomalous System Behavior on Control Device</i>
Observable 11 †	<i>Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC)</i>
Observable 12	Anomalous System Behavior on Control Device: Programmable Logic Controller (PLC): Anomalous Allocation of Memory Space

Observables Associated with the Manipulation of Control Technique (T0831)	
Observable 13	Anomalous Host Activity
Observable 14 †	<i>Anomalous Host Activity: Controlled Process in Anomalous State</i>
Observable 15 †	<i>Anomalous Host Activity: Industrial Systems Unresponsive</i>
Observable 16 †	<i>Anomalous Host Activity: Industrial Systems Unresponsive: Program Uploads Failed</i>
Observable 17 †	<i>Anomalous Host Activity: Industrial Systems Unresponsive: Inability to Enable Program Mode</i>
Observable 18 †	<i>Anomalous Host Activity: Anomalous Behavior of Servo Drives</i>
Observable 19	Anomalous Network Traffic: From Client Server to Local Hosts: Client Server with Python Executable to Open Communication Platform (OPC) Unified Architecture (UA) Hosts: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Write Request"
Observable 20	Anomalous Network Traffic: From Local Hosts to Client Server: From Open Communication Platform (OPC) Unified Architecture (UA) from Hosts Client Server with Python Executable: Over Open Platform Communications (OPC) Unified Architecture (UA) TCP Port 4840: Content contains "Write Response"
Observable 21 †	<i>Anomalous Host Activity: Local Devices Connected to Client Server: Open Communication Platform (OPC) Unified Architecture (UA) Device Associated with Local Open Communication Platform (OPC) Unified Architecture (UA) Client Server: Client Requiring Repeated Authentication</i>
Observable 22 †	<i>Anomalous Host Activity: Local Devices Connected to Client Server: Open Communication Platform (OPC) Unified Architecture (UA) Device Associated with Local Open Communication Platform (OPC) Unified Architecture (UA) Client Server: Device Unreachable Notification</i>

Observables Associated with the Damage to Property Technique (T0879)	
Observable 1 †	Anomalous Damage to Property

Observables Associated with the Loss of Safety Technique (T0880)	
Observable 1	Anomalous Host Activity
Observable 2 †	<i>Anomalous Host Activity: Industrial Systems Unresponsive</i>
Observable 3 †	<i>Anomalous Host Activity: Sustained Loss of Control</i>
Observable 4 †	<i>Anomalous Host Activity: Sustained Loss of Control: Industrial Control System Device</i>

[Copy and paste a new table for the observables for each technique, with a single carriage return (Normal style) between them for a little white space]

[Create a bookmark for the header in each observable table, then link the "Appendix A" text from the technique section to the matching bookmark in Appendix A.]

APPENDIX B: ARTIFACTS LIBRARY

Artifact	Description
Artifact	Description
Artifact	Description
Artifact	Description

Artifacts Associated with the Remote Services Technique (T0886)	
Artifact 1	Remote client connection
Artifact 2	logon event
Artifact 3	logoff
Artifact 4	logoff event
Artifact 5	registry changes
Artifact 6	registry connection change
Artifact 7	mouse movement
Artifact 8	unexpected I/O
Artifact 9	desktop prompt windows created
Artifact 10	session cache
Artifact 11	application log
Artifact 12	RDP traffic 3389
Artifact 13	system log event
Artifact 14	authentication logs
Artifact 15	GUI modifications
Artifact 16	Data file size in network content
Artifact 17	File movement
Artifact 18	MSSQL Traffic 1433 port
Artifact 19	SSH traffic 22
Artifact 20	SMB traffic 139, 445
Artifact 21	VNC traffic 5800, 5900
Artifact 22	process creation
Artifact 23	Remote Session creation timestamp

Artifacts Associated with the Remote Services Technique (T0886)

Artifact 24	network traffic content creation
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Artifacts Associated with the User Execution Technique (T0863)

Artifact 1	Application Log
Artifact 2	prefetch files
Artifact 3	system log
Artifact 4	registry modification
Artifact 5	file modifications
Artifact 6	file renaming
Artifact 7	system patches installed
Artifact 8	files opening
Artifact 9	file signature validation
Artifact 10	installers created
Artifact 11	process termination
Artifact 12	File Creation
Artifact 13	service termination
Artifact 14	file changes
Artifact 15	user account modification
Artifact 16	increased ICMP traffic (network scanning)
Artifact 17	File Execution
Artifact 18	Network Traffic changes
Artifact 19	Process Creation
Artifact 20	Network Connection creation
Artifact 21	Command execution
Artifact 22	application log content
Artifact 23	application installation

Artifacts Associated with the Command Line Execution Technique (T0807)

Artifact 1	Remote connections
Artifact 2	Script execution
Artifact 3	User account logon
Artifact 4	External network connection
Artifact 5	User account privilege change
Artifact 6	Command execution

Artifacts Associated with the Command Line Execution Technique (T0807)	
Artifact 7	Log on event ID 4624
Artifact 8	Event log type 3
Artifact 9	Event log type 10
Artifact 10	Failed Log on Event ID 4625
Artifact 11	cmd.exe application execution
Artifact 12	RDP traffic
Artifact 13	industrial application execution
Artifact 14	Powershell cmdlet application execution
Artifact 15	Event ID 4103 Powershell Command
Artifact 16	Event ID 4688 Command line execution
Artifact 17	NTUSER Application Execution Entries
Artifact 18	Command line memory data
Artifact 19	VNC traffic Port 5900
Artifact 20	SSH traffic
Artifact 21	Telnet traffic
Artifact 22	HTTP traffic
Artifact 23	Application Log
Artifact 24	Process creation
Artifact 25	Process ending

Artifacts Associated with the Native API Technique (T0834)	
Artifact 1	Industrial network traffic
Artifact 2	Industrial protocol command packet
Artifact 3	Device reads
Artifact 4	device I/O image table manipulated
Artifact 5	decive failure
Artifact 6	alter process logic
Artifact 7	device performance degradation
Artifact 8	device memory modification
Artifact 9	device alarm
Artifact 10	device live data changes
Artifact 11	Systems Calls
Artifact 12	alert generated
Artifact 13	memory corruption

Artifacts Associated with the Native API Technique (T0834)	
Artifact 14	host device failure
Artifact 15	blue screen
Artifact 16	performance degradation
Artifact 17	sysmon events created
Artifact 18	Services initiated
Artifact 19	Processes initiated
Artifact 20	Files created
Artifact 21	Imports hash changed
Artifact 22	.dll modifications
Artifact 23	System Resource Usage Management changes
Artifact 24	command execution
Artifact 25	configuration change

Artifacts Associated with the Change Operating Mode Technique (T0858)	
Artifact 1	Common Network Traffic
Artifact 2	end-point IP address
Artifact 3	application modification
Artifact 4	network connection reconfigured
Artifact 5	controlled device failure
Artifact 6	mode key position change*
Artifact 7	controlled device reboot
Artifact 8	controlled device manipulation
Artifact 9	log on event
Artifact 10	log off event
Artifact 11	user account information
Artifact 12	Industrial Network Traffic
Artifact 13	vendor proprietary protocol use
Artifact 14	network packet metadata change
Artifact 15	controlled device alarm
Artifact 16	process state change
Artifact 17	controlled device state change
Artifact 18	program file manipulation
Artifact 19	Remote Network Traffic
Artifact 20	Wireless connections

Artifacts Associated with the Change Operating Mode Technique (T0858)

Artifact 21	Radio connections
Artifact 22	.dll file creation
Artifact 23	application log event
Artifact 24	sender IP address

Artifacts Associated with the Modify Program Technique (T0889)

Artifact 1	Unexpected program download observed on network
Artifact 2	Modification to application responsible for program downloads
Artifact 3	Unexpected modification to program organizational units on a device

Artifacts Associated with the Valid Accounts Technique (T0859)

Artifact 1	Logons
Artifact 2	Default Credential Use
Artifact 3	Application Log
Artifact 4	Domain Permission Requests
Artifact 5	Permission Elevation Requests
Artifact 6	Application Use Times
Artifact 7	Configuration Changes
Artifact 8	Prefetch Files Created After Execution
Artifact 9	Logon Session Creation
Artifact 10	User Account Creation
Artifact 11	Authentication Creation
Artifact 12	System Logs
Artifact 13	Successful Logon Event ID 4624
Artifact 14	Failed Logons Event ID 4625
Artifact 15	Logon Timestamp
Artifact 16	Logon Type Entry

Artifacts Associated with the Exploitation for Privilege Escalation Technique (T0890)

Artifact 1	Unexpected process crash
Artifact 2	Network traffic associated with privilege escalation vulnerabilities (CVE-2014-4076 sent a specially crafted TCP packet to \\.\ TCP device through DeviceIoControl function)
Artifact 3	Unusual process activity (thread suspension of everything except thread running in a process other than exploit thread for CVE-2016-7255)

Artifacts Associated with the Exploitation for Privilege Escalation Technique (T0890)	
Artifact 4	Sysmon event 8 CreateRemoteThread process injection detected
Artifact 5	Unusual command line history associated with known CVE techniques (CVE-2019-5736 priv esc is visible via unusual command line commands)
Artifact 6	Suspicious file write to system directory followed by privileged execution of file
Artifact 7	Execution of a suspicious file in the system32 or windows directory at privileged level
Artifact 8	Unusual or unexpected kerberos ticket requests (CVE-2014-6423)
Artifact 9	Suspicious files written to disk
Artifact 10	Suspicious program running under SYSTEM or other elevated account
Artifact 11	Driver loaded (Sysmon Event ID 6)
Artifact 12	Network traffic matching vulnerability (snort, Suricata,
Artifact 13	Abnormal reads/writes between processes
Artifact 14	Unusual command line arguments to application (lolbins)
Artifact 15	Artifacts associated with known privilege escalation CVEs (PE hard coded debug file path for APT28 malware included reference to CVE-2014-4076 priv esc CVE)
Artifact 16	Unusual or unexpected child process running at elevated privileges

Artifacts Associated with the Change Operating Mode Technique (T0858)	
Artifact 1	Common Network Traffic
Artifact 2	end-point IP address
Artifact 3	application modification
Artifact 4	network connection reconfigured
Artifact 5	controlled device failure
Artifact 6	mode key position change*
Artifact 7	controlled device reboot
Artifact 8	controlled device manipulation
Artifact 9	log on event
Artifact 10	log off event
Artifact 11	user account information
Artifact 12	Industrial Network Traffic
Artifact 13	vendor proprietary protocol use
Artifact 14	network packet metadata change
Artifact 15	controlled device alarm
Artifact 16	process state change
Artifact 17	controlled device state change

Artifacts Associated with the Change Operating Mode Technique (T0858)

Artifact 18	program file manipulation
Artifact 19	Remote Network Traffic
Artifact 20	Wireless connections
Artifact 21	Radio connections
Artifact 22	.dll file creation
Artifact 23	application log event
Artifact 24	sender IP address

Artifacts Associated with the Rootkit Technique (T0851)

Artifact 1	Drive modification
Artifact 2	industrial network traffic for Remote Code Execution
Artifact 3	remote network traffic for Remote Code Execution
Artifact 4	Modified reporting
Artifact 5	Failed reporting
Artifact 6	Blue screen on workstation
Artifact 7	SRUM metadata changes
Artifact 8	performance mismatched with performance reporting
Artifact 9	Packet mismatch between network and host sources
Artifact 10	unusual protocol field values
Artifact 11	increased maintenance without faults reported
Artifact 12	Firmware number modification
Artifact 13	Blocked read requests at OS level
Artifact 14	Blocked delete requests at OS level
Artifact 15	Blocked create requests at OS level
Artifact 16	Blocked write requests at OS level
Artifact 17	Firmware flash update
Artifact 18	Device reboot
Artifact 19	Device failure
Artifact 20	Dropped API calls
Artifact 21	Application logs
Artifact 22	Dialog box requesting update
Artifact 23	common network traffic for Remote Code Execution

Artifacts Associated with the Remote System Discovery Technique (T0846)	
Artifact 1	Common Network Traffic
Artifact 2	IEC 103 Traffic (for North America)*
Artifact 3	IEC 61850 MMS and GOOSE*
Artifact 4	Controller Proprietary Traffic
Artifact 5	Echo Type 8 Traffic
Artifact 6	ICMP Type 7 Traffic
Artifact 7	SNMP Port 162 Traffic
Artifact 8	SNMP Port 161 Traffic
Artifact 9	Command Line dialog box open
Artifact 10	Operating system queries
Artifact 11	DNS port 53 zone transfers
Artifact 12	Industrial Network Traffic Content about Hostnames
Artifact 13	Polling network traffic from unauthorized IP Sender Addresses
Artifact 14	NetBIOS Name Services Port 137
Artifact 15	LDAP Port 389
Artifact 16	Active Directory calls
Artifact 17	Email server calls
Artifact 18	SMTP Port 25 traffic
Artifact 19	DNS lookup queries
Artifact 20	ARP scans
Artifact 21	TCP Connect Scan
Artifact 22	TCP SYN Scans
Artifact 23	Scans Over Industrial Network Ports with Target IPs
Artifact 24	TCP FIN Scans
Artifact 25	TCP Reverse Ident Scan
Artifact 26	TCP XMAS Scan
Artifact 27	TCP ACK Scan*
Artifact 28	VNC port 5900 calls
Artifact 29	Protocol content enumeration
Artifact 30	Protocol header enumeration
Artifact 31	Recurring protocol SYN traffic
Artifact 32	Sequential protocol SYN traffic
Artifact 33	Statistical anomalies in network traffic
Artifact 34	Industrial Network Traffic Content containing Logical Identifiers
Artifact 35	Device failure

Artifacts Associated with the Remote System Discovery Technique (T0846)	
Artifact 36	Device reboot
Artifact 37	Bandwidth degradation
Artifact 38	Host recent connection logs
Artifact 39	Industrial Network Traffic
Artifact 40	OPC Network Traffic
Artifact 41	IEC 104
Artifact 42	IEC 102
Artifact 43	IEC 101 Traffic to Serial Devices

Artifacts Associated with the Remote System Information Discovery Technique (T0888)	
Artifact 1	Unexpected Industrial protocol usage
Artifact 2	Unexpected Industrial application usage
Artifact 3	Unexpected standard protocol usage
Artifact 4	Unexpected recon associated command line options (ping sweep, netstat, etc)
Artifact 5	Unexpected recon associated child processes (ping sweep, netstat, etc)
Artifact 6	Unexpected recon associated library calls
Artifact 7	Exfiltration of host, network, and/or system architecture or configuration data
Artifact 8	Compromise and exfiltration of data from asset information datastores or applications

Artifacts Associated with the Network Sniffing Technique (T0842)	
Artifact 1	ARP Traffic
Artifact 2	Network Switch logons
Artifact 3	Network switch configuration changes
Artifact 4	DNS Traffic
Artifact 5	Traffic redirection
Artifact 6	New Send IP Addresses
Artifact 7	New Receiver Addresses
Artifact 8	All network traffic redirected
Artifact 9	Switch changed to promiscuous mode
Artifact 10	Switch mirror port opened
Artifact 11	Industrial protocol traffic redirected

Artifacts Associated with the Valid Accounts Technique (T0859)	
Artifact 1	Logons
Artifact 2	Default Credential Use
Artifact 3	Application Log
Artifact 4	Domain Permission Requests
Artifact 5	Permission Elevation Requests
Artifact 6	Application Use Times
Artifact 7	Configuration Changes
Artifact 8	Prefetch Files Created After Execution
Artifact 9	Logon Session Creation
Artifact 10	User Account Creation
Artifact 11	Authentication Creation
Artifact 12	System Logs
Artifact 13	Successful Logon Event ID 4624
Artifact 14	Failed Logons Event ID 4625
Artifact 15	Logon Timestamp
Artifact 16	Logon Type Entry

Artifacts Associated with the Default Credentials Technique (T0812)	
Artifact 1	
Artifact 2	
Artifact 3	
Artifact 4	
Artifact 5	
Artifact 6	
Artifact 7	
Artifact 8	
Artifact 9	
Artifact 10	
Artifact 11	
Artifact 12	
Artifact 13	
Artifact 14	
Artifact 15	
Artifact 16	

Artifacts Associated with the Default Credentials Technique (T0812)	
Artifact 17	
Artifact 18	
Artifact 19	
Artifact 20	
Artifact 21	
Artifact 22	
Artifact 23	
Artifact 24	
Artifact 25	

Artifacts Associated with the Program Download Technique (T0843)	
Artifact 1	Controller In Stop State
Artifact 2	Operational Process Restart
Artifact 3	Operational Database Data Modification
Artifact 4	Device Alert
Artifact 5	Device Alarm
Artifact 6	Controller Application Log Event
Artifact 7	Supervisory Workstation Program Download Popup
Artifact 8	Controller Application Log Type
Artifact 9	Controller Application Log Timestamp
Artifact 10	Controller Network Connections via Management Protocol
Artifact 11	Controller Connection to External Website
Artifact 12	Controller in Program State
Artifact 13	Controller Connected to External Networks
Artifact 14	Network Traffic Creation
Artifact 15	Network Metadata
Artifact 16	External Domain Connection
Artifact 17	External IP Address
Artifact 18	Controller State Change
Artifact 19	Operational Process Shutdown

Artifacts Associated with the Lateral Tool Transfer Technique (T0867)	
Artifact 1	command execution
Artifact 2	file location change

Artifacts Associated with the Lateral Tool Transfer Technique (T0867)	
Artifact 3	file metadata changes
Artifact 4	user information changes
Artifact 5	process creation
Artifact 6	System Resource Usage Management events
Artifact 7	Data sent from one location to another
Artifact 8	Data received from one location to another
Artifact 9	SQL commands
Artifact 10	SQL create commands
Artifact 11	SQL insert commands
Artifact 12	command prompt dialog box open
Artifact 13	SMB traffic
Artifact 14	.dll injection into file directory
Artifact 15	.dll execution
Artifact 16	powershell dialog box open
Artifact 17	common network traffic
Artifact 18	remote network traffic
Artifact 19	industrial network traffic
Artifact 20	file creation
Artifact 21	file modification
Artifact 22	file deletion

Artifacts Associated with the Remote Services Technique (T0886)	
Artifact 1	Remote client connection
Artifact 2	logon event
Artifact 3	logoff
Artifact 4	logoff event
Artifact 5	registry changes
Artifact 6	registry connection change
Artifact 7	mouse movement
Artifact 8	unexpected I/O
Artifact 9	desktop prompt windows created
Artifact 10	session cache
Artifact 11	application log
Artifact 12	RDP traffic 3389

Artifacts Associated with the Remote Services Technique (T0886)	
Artifact 13	system log event
Artifact 14	authentication logs
Artifact 15	GUI modifications
Artifact 16	Data file size in network content
Artifact 17	File movement
Artifact 18	MSSQL Traffic 1433 port
Artifact 19	SSH traffic 22
Artifact 20	SMB traffic 139, 445
Artifact 21	VNC traffic 5800, 5900
Artifact 22	process creation
Artifact 23	Remote Session creation timestamp
Artifact 24	network traffic content creation

Artifacts Associated with the Point and Tag Identification Technique (T0)	
Artifact 1	Industrial Network Traffic
Artifact 2	control server log off
Artifact 3	application logs
Artifact 4	application manipulation
Artifact 5	application user event
Artifact 6	application reads
Artifact 7	application copy
Artifact 8	point and tag data exfiltration
Artifact 9	host system registry modification
Artifact 10	user registry changes
Artifact 11	memory location changes
Artifact 12	Common Network traffic
Artifact 13	OPC requests
Artifact 14	.dll creation
Artifact 15	.dll execution
Artifact 16	.dll hooking
Artifact 17	SQL network traffic
Artifact 18	Database vendor specific protocol request
Artifact 19	network traffic content focused on point and tag reads
Artifact 20	External point and tag read requests over network trust boundaries

Artifacts Associated with the Point and Tag Identification Technique (T0)	
Artifact 21	Data historian reads
Artifact 22	Data historian writes
Artifact 23	Database reads
Artifact 24	Control server reads
Artifact 25	Data historian log on event
Artifact 26	database log on event
Artifact 27	control server log on
Artifact 28	DNS Queries Traffic Port 53
Artifact 29	MAC Address
Artifact 30	Source IP Address
Artifact 31	Destination IP Address
Artifact 32	Network discover protocols
Artifact 33	SMB Port 445
Artifact 34	SNMP Port 161
Artifact 35	LLDP Requests
Artifact 36	HTTP Port 80
Artifact 37	Ping Echo Port 7 (ICMP)
Artifact 38	Static Source IP Address
Artifact 39	Usage of default account
Artifact 40	Usage of vendor maintenance account
Artifact 41	Domain name
Artifact 42	Domain registrant data
Artifact 43	Domain IP resolution
Artifact 44	Domain Autonomous System Number (ASN)
Artifact 45	Hardware serial number missing
Artifact 46	Additional hardware inserted on devices
Artifact 47	Mismatched software hashes
Artifact 48	Device failure
Artifact 49	Device incompatibility issues
Artifact 50	Hardware tampering evidence
Artifact 51	Hardware failed site acceptance test
Artifact 52	Physical defects to hardware
Artifact 53	Unscheduled firmware updates

Artifacts Associated with the Monitor Process State Technique (T0801)	
Artifact 1	Network Read Request
Artifact 2	OPC Read Requests
Artifact 3	Local memory read requests
Artifact 4	Database Read Request
Artifact 5	External Network connections
Artifact 6	Internal Network Connections
Artifact 7	IP Addresses
Artifact 8	MAC Addresses
Artifact 9	Operational data exfiltration
Artifact 10	User Account Creation
Artifact 11	User Account Privilege Change
Artifact 12	SQL Read Requests

Artifacts Associated with the Program Upload Technique (T0845)	
Artifact 1	Common network traffic protocols
Artifact 2	Application dialog box open
Artifact 3	Log on event
Artifact 4	Log off event
Artifact 5	Application authentication event
Artifact 6	Application metadata reported
Artifact 7	Industrial network traffic
Artifact 8	Product specific protocols
Artifact 9	Data sent to unknown IP Addresses
Artifact 10	Controller sending data to new IP Address
Artifact 11	Application log
Artifact 12	Application log event content
Artifact 13	Large data file transfers
Artifact 14	Vendor software used at strange time

Artifacts Associated with the Detect Operating Mode Technique (T0868)	
Artifact 1	Industrial Network Traffic
Artifact 2	Remote Network traffic
Artifact 3	Vendor Specific Protocol Traffic
Artifact 4	Controller reads

Artifacts Associated with the Detect Operating Mode Technique (T0868)

Artifact 5	Controller writes
Artifact 6	Engineering Workstation project file reads
Artifact 7	HMI end point connections with operating mode content
Artifact 8	External operating mode read requests over network trust boundaries

Artifacts Associated with the Technique (T0869)

Artifact 1	External Network Connections
Artifact 2	DNS Autonomous System Number
Artifact 3	Increase in the number of external connections
Artifact 4	Network Content Metadata
Artifact 5	Network Connection Times
Artifact 6	HTTP Traffic Port 80
Artifact 7	DNS Traffic Port 53
Artifact 8	SMB Traffic Port 445
Artifact 9	HTTPS Traffic Port 443
Artifact 10	RDP Traffic Port 3389
Artifact 11	HTTP Post Request
Artifact 12	External IP Addresses

Artifacts Associated with the Commonly Used Port Technique (T0885)

Artifact 1	Unexpected process usage of common port observed via OS commands (netstat)
Artifact 2	Unexpected process usage of common port observed via memory
Artifact 3	Unexpected process usage of common port observed via OS logs
Artifact 4	Unexpected process usage of common port observed via firewall logs
Artifact 5	Unexpected host communicating with common port on industrial asset

Artifacts Associated with the Denial of Service Technique (T0814)

Artifact 1	Application Log
Artifact 2	TDS Traffic Increase port 1433
Artifact 3	Increase Industrial Protocol Exceptions
Artifact 4	Low Resources Warning
Artifact 5	Ransom notice
Artifact 6	Services failure

Artifacts Associated with the Denial of Service Technique (T0814)	
Artifact 7	Network Traffic Connection Increase
Artifact 8	IP Addresses
Artifact 9	MAC Addresses
Artifact 10	External Network Connections
Artifact 11	Process performance degrades
Artifact 12	Operational data corruption
Artifact 13	Application failure
Artifact 14	ICMP Echo port 7 Traffic Increase

Artifacts Associated with the Data Destruction Technique (T0809)	
Artifact 1	Program execution
Artifact 2	Telnet Port 23
Artifact 3	SFTP Port 22
Artifact 4	FTPS Port 990
Artifact 5	SMB Port 139, 445
Artifact 6	HTTP Port 80
Artifact 7	HTTPS Port 443
Artifact 8	Command line arguments
Artifact 9	SCP Port 22
Artifact 10	Memory corruption
Artifact 11	Files moved to Recycle Bin
Artifact 12	Non-Native Files
Artifact 13	Transient device connections
Artifact 14	External network connections
Artifact 15	Local network connections
Artifact 16	Host system reboot failure
Artifact 17	Process logic failure
Artifact 18	Event log creation
Artifact 19	System call
Artifact 20	System application interruption
Artifact 21	Device failure
Artifact 22	Recovery Attempt failure
Artifact 23	File encryptions
Artifact 24	Missing files

Artifacts Associated with the Data Destruction Technique (T0809)

Artifact 25	Use of file transfer protocols
Artifact 26	FTP Port 20, 21
Artifact 27	TFTP Port 60

Artifacts Associated with the Service Stop Technique (T0881)

Artifact 1	Process failure
Artifact 2	Alarm Event
Artifact 3	Internal System Logs
Artifact 4	Application Error Messages
Artifact 5	Process Error Messages
Artifact 6	Application service stop
Artifact 7	OS Service Crash
Artifact 8	System Event Logs
Artifact 9	Application Event Logs
Artifact 10	OS API call
Artifact 11	Command line system argument
Artifact 12	System Resource Usage Manager application usage change
Artifact 13	Registry Change HKLM\System\CurrentControlSet\Services

Artifacts Associated with the Modify Parameter Technique (T0836)

Artifact 1	network traffic
Artifact 2	alert failure
Artifact 3	project file changes
Artifact 4	driver modifications
Artifact 5	remote GUI manipulation
Artifact 6	.dll changes
Artifact 7	process creation
Artifact 8	nonstandard service creation
Artifact 9	new software installed
Artifact 10	non standard application calls
Artifact 11	industrial protocol packets
Artifact 12	device set points changed
Artifact 13	device alert
Artifact 14	device failure

Artifacts Associated with the Modify Parameter Technique (T0836)

Artifact 15	process performance degradation
Artifact 16	application log events
Artifact 17	configuration changes
Artifact 18	machine state change

Artifacts Associated with the Unauthorized Command Message Technique (T0855)

Artifact 1	Protocol Specific Command Packet
Artifact 2	Machine State Change
Artifact 3	Process restart
Artifact 4	Process failure
Artifact 5	Network resets
Artifact 6	Protocol Metadata Change
Artifact 7	Process Timing Change
Artifact 8	Process Logic Change
Artifact 9	MAC Addresses
Artifact 10	IP Addresses
Artifact 11	Operational Application Log
Artifact 12	Operational Data Created
Artifact 13	Process Alarm
Artifact 14	Process Alarm Event
Artifact 15	Application Level I/O Manipulation
Artifact 16	OS Level I/O Manipulation

Artifacts Associated with the Device Shutdown or Restart Technique (T0816)

Artifact 1	Process Application Event
Artifact 2	Process environmental changes
Artifact 3	Loss of network connection
Artifact 4	Network Command packets
Artifact 5	Reboot screen
Artifact 6	Blue screen
Artifact 7	Significant operational data changes
Artifact 8	Log on events
Artifact 9	Log off events
Artifact 10	Process Failure

Artifacts Associated with the Device Shutdown or Restart Technique (T0816)

Artifact 11	Hardware Failure
Artifact 12	Command Prompt Opened
Artifact 13	Unauthorized Input
Artifact 14	Memory corruption
Artifact 15	Process alarm
Artifact 16	External network connections
Artifact 17	Local network connections

Artifacts Associated with the Theft of Operational Information Technique (T0882)

Artifact 1	Exfiltration of endpoint host data (spreadsheets, diagrams, documents, configurations, etc) via industrial protocols
Artifact 2	Exfiltration of endpoint host data (spreadsheets, diagrams, documents, configurations, etc) via standard protocols
Artifact 3	Exfiltration from database via standard queries
Artifact 4	Exfiltration of operational info via phishing

Artifacts Associated with the Denial of Control Technique (T0813)

Artifact 1	Input failure
Artifact 2	Increased network packet delivery
Artifact 3	Process failure
Artifact 4	Process reboot
Artifact 5	Serial communication failure
Artifact 6	Network ports opened
Artifact 7	Network ports closed
Artifact 8	Process nonresponsive

Artifacts Associated with the Denial of View Technique (T)

Artifact 1	Application logic hooks or modifications might prevent proper reporting to/from an industrial application (Stuxnet DLL hooking)
Artifact 2	Blocking communications paths and channels (comms, network infrastructure, host packet routing) might occur via an OS or device modification
Artifact 3	Masked reconfiguration of alarm thresholds or other configuration settings might result in unexpected lack of situational awareness
Artifact 4	File system modification artifacts might be associated with the denial of view might be present on disk

Artifacts Associated with the Loss of Availability Technique (T0826)	
Artifact 1	Operator or user discovery of encrypted or inoperable systems
Artifact 2	Significant logged usage of native crypto functions or presence of import of crypto functions in binaries
Artifact 3	Significant reduction or increase in network traffic due to malware propagation of disappearance of services
Artifact 4	Unexplained loss of application data
Artifact 5	Unexplained loss of user data
Artifact 6	Process failure due to loss of required network or system dependency
Artifact 7	Changes in network routing or usage of redundant control system network connection due to failed network path
Artifact 8	File system modification artifacts might be associated with the loss of availability might be present on disk

Artifacts Associated with the Manipulation of Control Technique (T0831)	
Artifact 1	HMI Input manipulation
Artifact 2	Command Execution
Artifact 3	Event Log Creation
Artifact 4	Application Log Event
Artifact 5	Altered Command Sequences
Artifact 6	Application File Modification
Artifact 7	Operational Data Modification
Artifact 8	I/O Modification
Artifact 9	Engineering Workstation Mouse Movement
Artifact 10	Controller Set Point Change
Artifact 11	Controller Parameter Change
Artifact 12	Controller Tag Change
Artifact 13	Process State Change
Artifact 14	Process Shutdown
Artifact 15	Process Restart
Artifact 16	Process Initiated

Artifacts Associated with the Damage to Property Technique (T0879)	
Artifact 1	Damage to property due to equipment malfunction
Artifact 2	frequent maintenance failures
Artifact 3	process trip
Artifact 4	alarms

Artifacts Associated with the Damage to Property Technique (T0879)	
Artifact 5	smoke
Artifact 6	liquid spills
Artifact 7	pressure relief
Artifact 8	loud vibrations
Artifact 9	safety systems engaged
Artifact 10	breakers closing and opening rapidly
Artifact 11	Damage to property due to malicious network traffic
Artifact 12	Damage to property due to equipment degradation
Artifact 13	Reduction in traffic volume to device
Artifact 14	Increase in connecting error to device
Artifact 15	Catastrophic failure
Artifact 16	Surges in power
Artifact 17	ladder logic configuration changes
Artifact 18	industrial network traffic

Artifacts Associated with the Technique (T0880)	
Artifact 1	Loss of access to a safety system
Artifact 2	Loss of control of a safety system
Artifact 3	Malicious firmware update to a safety system

[Copy and paste a new table for the observables for each technique, with a single carriage return (Normal style) between them for a little white space]

Contact database administrator with techniques used, and they will give you the artifact table content and count.

APPENDIX C: OBSERVERS

This is a collection of standardized potential observers that work in operational technology organizations. It has been slightly modified by the CyOTE team from the [Job Role Groupings listed in the SANS ICS Job Role to Competency Level Poster](#) to communicate the categories of potential observers during cyber events.

Engineering  • Process Engineer • Electrical, Controls, and Mechanical Engineer • Project Engineer • Systems and Reliability Engineer • OT Developer • PLC Programmer • Emergency Operations Manager • Plant Networking • Control/Instrumentation Specialist • Protection and Controls • Field Engineer • System Integrator	Support Staff  • Remote Maintenance & Technical Support • Contractors (engineering) • IT and Physical Security Contractor • Procurement Specialist • Legal • Contracting Engineer • Insurance • Supply-chain Participant • Inventory Management/Lifecycle Management • Physical Security Specialist
Operations Technology (OT) Staff  • Operator • Site Security POC • Technical Specialists (electrical/mechanical/chemical) • ICS/SCADA Programmer	Information Technology (IT) Cybersecurity  • ICS Security Analyst • Security Engineering and Architect • Security Operations • Security Response and Forensics • Security Management (CSO) • Audit Specialist • Security Tester
Operational Technology (OT) Cybersecurity  • OT Security • ICS/SCADA Security	
Management  • Plant Manager • Risk/Safety Manager • Business Unit Management • C-level Management	Information Technology (IT) Staff  • Networking and Infrastructure • Host Administrator • Database Administrator • Application Development • ERP/MES Administrator • IT Management

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² [Dragos | "CHERNOVITE's PIPEDREAM Malware Targeting Industrial Control Systems (ICS)" | <https://www.dragos.com/blog/industry-news/chernovite-pipedream-malware-targeting-industrial-control-systems/> | 13 April 2022 | Accessed on 22 March 2022 | The source is publicly available information and does not contain classification markings]

³ [S4 Events | Rob M Lee | "PIPEDREAM - Most Flexible and Capable ICS Malware to Date" | <https://www.youtube.com/watch?v=H82sblwFxt4> | 2 May 2022 | Accessed on 15 November 2022 | Source is publicly available information and does not contain classification markings]

⁴ [S4 Events | Nathan Brubaker & Keith Lunden | "INCONTROLLER" | <https://www.youtube.com/watch?v=ro7qo0iZfgw> | 2 May 2022 | Accessed on 15 November 2022 | The source is publicly available information and does not contain classification markings]

⁵ [Mandiant | Brubaker & Others | "INCONTROLLER: New State-Sponsored Cyber Attack Tools Target Multiple Industrial Control Systems" | <https://www.mandiant.com/resources/blog/incontroller-state-sponsored-ics-tool> | 13 April 2022 | Accessed on 14 November 2022 | The source is publicly available information and does not contain classification markings]

⁶ [Mandiant | Brubaker & Others | "INCONTROLLER: New State-Sponsored Cyber Attack Tools Target Multiple Industrial Control Systems" | <https://www.mandiant.com/resources/blog/incontroller-state-sponsored-ics-tool> | 13 April 2022 | Accessed on 14 November 2022 | The source is publicly available information and does not contain classification markings]

⁷ [Mandiant | Brubaker & Others | "INCONTROLLER: New State-Sponsored Cyber Attack Tools Target Multiple Industrial Control Systems" | <https://www.mandiant.com/resources/blog/incontroller-state-sponsored-ics-tool> | 13 April 2022 | Accessed on 14 November 2022 | The source is publicly available information and does not contain classification markings]

⁸ {SANS| Chad Tilbury | "Investigating WMI Attacks" | <https://www.sans.org/blog/investigating-wmi-attacks/> | 9 February 2019 | Accessed on 14 November 2022 | The source is publicly available information and does not contain classification markings]

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¹¹ [Mandiant | Brubaker & Others | "INCONTROLLER: New State-Sponsored Cyber Attack Tools Target Multiple Industrial Control Systems" | <https://www.mandiant.com/resources/blog/incontroller-state-sponsored-ics-tool> | 13 April 2022 | Accessed on 14 November 2022 | The source is publicly available information and does not contain classification markings]

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- ¹³ [DEF CON | Jimmy Wylie | “Analyzing PIPEDREAM - Challenges in Testing an ICS Attack Toolkit” | <https://www.youtube.com/watch?v=dz6VNYSSJ0> | 20 Oct 2022 | Accessed on 3 February 2023 | Source is publicly available information and does not contain classification markings]
- ¹⁴ [DEF CON | Jimmy Wylie | “Analyzing PIPEDREAM - Challenges in Testing an ICS Attack Toolkit” | <https://www.youtube.com/watch?v=dz6VNYSSJ0> | 20 Oct 2022 | Accessed on 3 February 2023 | Source is publicly available information and does not contain classification markings]
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- ¹⁷ Dragos | “CHERNOVITE’s PIPEDREAM Malware Targeting Industrial Control Systems (ICS)” | <https://www.dragos.com/blog/industry-news/chernovite-pipedream-malware-targeting-industrial-control-systems/> | 13 April 2022 | Accessed on 22 March 2022 | The source is publicly available information and does not contain classification markings]
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²⁸ ²⁸ [Mandiant | Brubaker & Others | “INCONTROLLER: New State-Sponsored Cyber Attack Tools Target Multiple Industrial Control Systems” | <https://www.mandiant.com/resources/blog/incontroller-state-sponsored-ics-tool> | 13 April 2022 | Accessed on 14 November 2022 | The source is publicly available information and does not contain classification markings]

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